



# Open access battle damage detection via Pixel-Wise T-Test on Sentinel-1 imagery

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## ABSTRACT

In the context of recent, highly destructive conflicts in Gaza and Ukraine, reliable estimates of building damage are essential for an informed public discourse, human rights monitoring, and humanitarian aid provision. Given the contentious nature of conflict damage assessment, these estimates must be fully **reproducible**, explainable, and derived from open access data. This paper introduces a new method for building damage detection—the Pixel-Wise T-Test (PWTT)—that satisfies these conditions. Using a combination of freely-available synthetic aperture radar imagery and statistical change detection, the PWTT generates accurate conflict damage estimates across a wide area at regular time intervals. Accuracy is assessed using an original dataset of over 2 million labeled building footprints spanning 30 cities across Palestine, Ukraine, Sudan, Syria, and Iraq. Despite being simple and lightweight, the algorithm achieves building-level accuracy statistics (AUC=0.87 in the full sample) rivaling state of the art methods that use deep learning and high resolution imagery. The workflow is **open source** and deployed entirely within the Google Earth Engine environment, allowing for the generation of interactive Battle Damage Dashboards for **Ukraine** and **Gaza** that update in near-real time, enabling the public and humanitarian practitioners to immediately get estimates of damaged buildings in a given area.

## 1. Introduction

The accurate and timely identification of buildings damaged by war is crucial for local civilians, aid agencies, and an informed public. Recent progress has been made in automatically identifying damaged buildings using deep learning and very-high resolution (VHR) optical satellite imagery (Shen et al., 2022; Xia et al., 2023; Zheng et al., 2021; Mueller et al., 2021; Lee et al., 2020). However, these models often struggle to generalize and are both financially and computationally expensive. As such, there has been little uptake of these methods by humanitarian practitioners such as the United Nations Satellite Agency (UNOSAT), which continues to identify damaged buildings by manually combining through satellite imagery.

This paper develops a new algorithm for the detection of urban conflict damage – the Pixelwise T-Test (PWTT) – that is lightweight, unsupervised, and uses only freely-available Synthetic Aperture Radar (SAR) imagery. This addresses many of the problems associated with expense, coverage consistency, explainability, and domain shift that are inherent to the use of deep learning on optical imagery.

The recent, devastating conflicts in Gaza and Ukraine have evidenced significant public demand for open access conflict damage estimates; the PWTT was deployed by the Economist magazine to assess building damage across all of Ukraine at multiple time intervals in the first year of the conflict (Economist, 2023). Yet, there are no

studies that conduct large scale, building-level accuracy assessments of SAR-derived conflict damage detection.

As such, a second key contribution of this paper is the creation and release of a novel dataset combining manual building damage annotations conducted by the UN with 2,277,345 building footprints spanning 30 cities across Ukraine, Syria, and Iraq, Sudan as well as the Gaza Strip at seven time periods since October 2023. This is used to conduct comprehensive validation of the PWTT algorithm in different conflict settings. Results indicate that the PWTT can identify damaged buildings with accuracy rivaling that of state-of-the-art deep learning approaches that rely on expensive high-resolution imagery, achieving an Area Under the Curve of 0.87 for the full sample.

Finally, the lightweight and open-access nature of the PWTT enables the creation of dynamic Battle Damage Dashboards, updating with new imagery in near-real time, offering country-wide damage assessments across any war time frame. The dashboards integrate damage estimates with high-resolution pre-war population data, gauging the displaced population in destroyed areas. The dashboards also incorporate other open-source data, such as geolocated social media footage, providing supplementary verification. This simple, robust approach constitutes a significant advancement in real-time public monitoring of armed conflict.

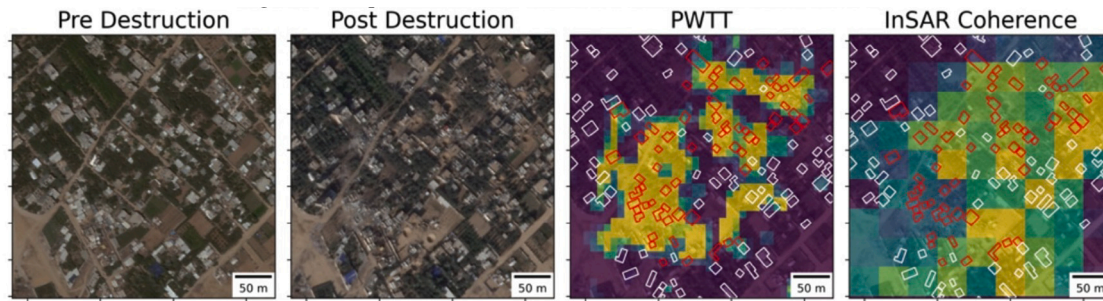
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**Fig. 1.** Comparison of PWTT and InSAR coherence resolution in Gaza.

The results of the Pixel-Wise T-Test are compared to Interferometric Coherence Change Detection. On the right two subplots, buildings labeled as damaged by UNOSAT are overlaid in red, while undamaged buildings are shown in white. The PWTT is able to maintain the sensor's native resolution of 10 meters per pixel, making 16 times higher resolution than traditional InSAR methods, thereby mitigating spatial spillovers in dense urban environments. Pre and Post-conflict images ©2024 Planet Labs PBC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

### 1.1. Building damage assessment using deep learning

Considerable progress has been made in the field of building damage assessment (BDA) using remote sensing. One of the most significant recent developments has been release of the xBD dataset, which consists of over 45,000 km<sup>2</sup> of high resolution satellite imagery and over 850,000 annotated building footprints from natural disasters around the world (Gupta et al., 2019). xBD has become the benchmark dataset for both training and evaluating deep learning-based building damage detection models, which have since proliferated (Shen et al., 2022; Xia et al., 2023; Deng and Wang, 2022). State of the art performance was achieved by Zheng et al. (2021), who introduced a deep object-based semantic change detection framework called ChangeOS which reached F1 accuracy of 0.75 for the task of building damage detection on the xBD dataset. However, no equivalent benchmark exists for conflict-induced building damage assessment and there are several domain-specific challenges that distinguish the latter from general building damage assessment from natural disasters.

Whereas natural disasters typically involve a single destructive event affecting a relatively small area, battle damage can gradually accrue over months or years across a very large area. Neural networks work well for the former, but are harder to implement for multi-temporal wide-area monitoring. One of the most influential recent studies of conflict-induced destruction is the Mueller et al. (2021) study of the Syrian civil war, which achieves high accuracy by training a Convolutional Neural Network on damage annotations carried out by the United Nations and high resolution satellite imagery. When training and testing on the same cities, they achieve an Area Under the Curve (AUC) of 0.9. Yet the model struggles to generalize even within Syria: when a model is only trained on Aleppo and inference is conducted on other Syrian cities, accuracy statistics drop by as much as 29%, with the average AUC for these cities falling to 0.7. Similarly, when Zheng et al. (2021) test a model trained using the xBD dataset on the 2020 Beirut explosion, F1 accuracy drops from 75% to 48%. A consistent feature in the application of deep learning to the task of building damage detection is significantly worse out-of-the-box performance in new geographies.

Compared to natural disasters, the contentious nature of battle damage assessments impose a set of additional requirements for verifiable impartiality. Results must be fully reproducible – requiring both code and necessary inputs to be open access – and explainable. Neural networks are often referred to as “black boxes”, and do not yield explainable results. Most deep learning approaches also eschew open data in favor of expensive, high resolution optical imagery; monitoring large areas can become very costly, particularly if repeated observations are required. At \$23/km<sup>2</sup> for new high-resolution satellite imagery from MAXAR (the kind used by Mueller et al. (2021), and in the xBD dataset), it would cost \$13.9 million to image the

entirety of Ukraine just once (Mueller et al., 2021; Gupta et al., 2019; AAAS, 2024). Clouds pose an additional challenge for optical imagery; while damage assessment for conflicts in the Middle East may not be significantly hampered by cloud cover, certain areas in Ukraine are often shrouded in clouds. As such, while much of the recent literature on building damage detection utilizes high resolution optical imagery and deep learning, these methods are less well suited to the study of conflict-induced damage.

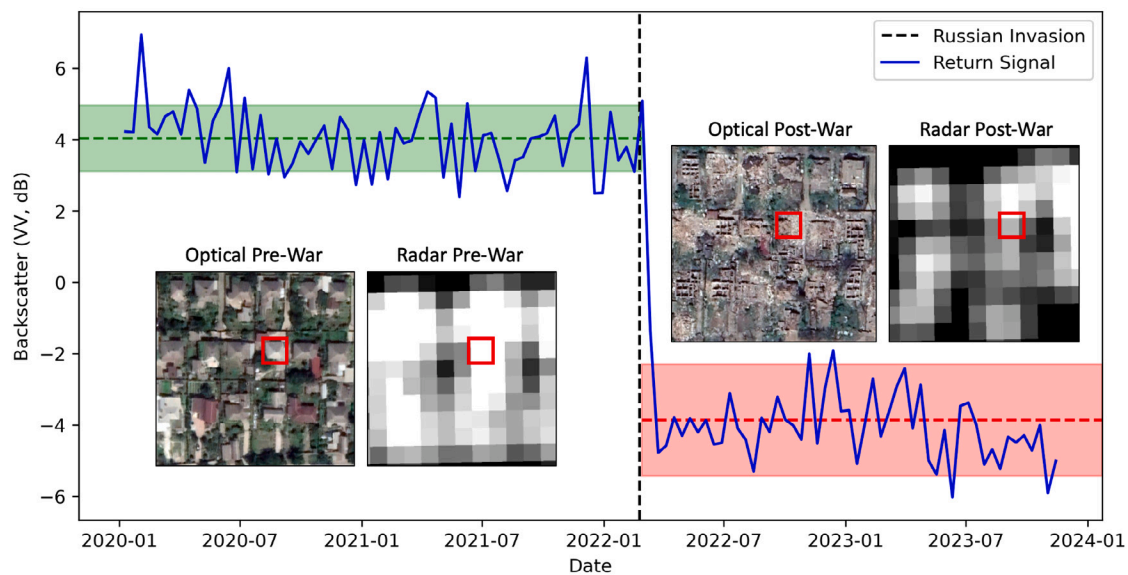
### 1.2. Building damage assessment in synthetic aperture radar imagery

Synthetic Aperture Radar (SAR) imagery has several properties that make it useful for building damage assessment, including weather- and illumination-independent images and a high repeat time (ESA, 2024). The SAR sensor emits a pulse of microwaves towards the earth and measures the return signal (backscatter), enabling the analysis of textural features on the earth's surface based on how they reflect the signal back to the satellite.

There is a vast literature on building damage detection using SAR imagery, largely in the field of natural disaster response. These are broadly divided into two categories, on the basis of whether they rely on detecting changes in the phase or the amplitude of the return signal. The approach taken herein conducts amplitude-based change detection.

The most common amplitude-based approach to building damage detection involves simply calculating the difference or ratio between one pre-disaster image and one post-disaster image (Liu and Yamazaki, 2011; Gokon et al., 2017; Dell'Acqua et al., 2010; Wang and Jin, 2009). A significant change in signal amplitude would have to result from a change in a building's scattering mechanism – for example the transition from double-bounce scattering to diffuse scattering as a standing building turns to rubble – rather than smaller changes (Ge et al., 2020). However, backscatter amplitude is affected by the dielectric properties of the target, such as the roofing material and the presence of water or snow which may reduce accuracy particularly if only a small number of images are used. Several studies have found that given only a pair of images, phase-based change detection outperforms amplitude-based methods particularly for the detection of minor damage (Liu and Yamazaki, 2017; Arciniegas et al., 2007)

As such, coherence change detection (CCD) is the more widely used method for building damage assessment in SAR imagery, and this approach has a number of advantages. Because it analyzes the shift in the returning signal's phase, CCD is able to pick up on minute changes on the ground owing to the fact that Sentinel-1 transmits C-band signals with a wavelength of just 5.6 centimeters (Ge et al., 2020). This sensitivity has made CCD particularly widespread in the analysis of earthquake damage, as it is not only capable of detecting damaged buildings but centimeter-level ground subsidence (Ito et al., 2000; Yonezawa and Takeuchi, 2001). However, this sensitivity can also be



**Fig. 2.** Time-series Sentinel-1 Backscatter amplitude of a damaged building in Mariupol, Ukraine.

Change in backscatter amplitude resulting from the destruction of a building in Mariupol, Ukraine, following the Russian invasion. The green dashed line and shaded area represent the pixel's mean backscatter amplitude  $\pm 1$  standard deviation prior to the invasion, while the red line and shaded area represent these statistics following the building's destruction. Sub-figures show optical imagery from Google Earth and radar imagery from Sentinel-1 of the area before and after destruction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a liability, as “coherent change detection methods tend to have very large false positive rates, where change is vastly overestimated.” Meyer (2019, p. 19). Furthermore, the image processing involved in the calculation of CCD results in a 16-fold reduction in resolution, from 10 meters per pixel to 40m/px. At this resolution, 98% of the building footprints in the validation dataset are smaller than an individual pixel, as shown in Fig. 1. High sensitivity and low resolution, particularly in the context of conflict in dense urban environments, create a significant risk of false positives.

## 2. Methods

The damage detection algorithm developed herein is a multi-temporal Pixel-Wise T-Test computed using backscatter amplitude, which mitigates many of the limitations of existing coherence and amplitude-based building damage detection methods. By relying on backscatter amplitude, it allows for the maintenance of the imagery's native resolution of 10 meters and is not overly sensitive to minor changes when compared with coherence change detection. By integrating pre- and post-conflict pixel standard deviation over extended time periods, it better distinguishes building damage from general anthropogenic change and seasonal variation that often affect amplitude-based methods. The computation of the test statistic is simple enough to be calculated by hand, and can consequently be applied over very large areas at regular time intervals. The use of time-series statistical change detection obviates the need for training and fine-tuning, and mitigates many of the problems associated with overfitting and domain shift that plague neural networks.

### 2.1. Data

#### 2.1.1. Synthetic aperture radar imagery

The main source of data used in this study is Synthetic Aperture Radar (SAR) imagery from the European Space Agency's Sentinel-1 satellite. Sentinel-1 is a dual polarimetric SAR system, emitting C-band radar microwaves with a vertical polarization and recording both vertical and horizontal reflected polarizations (ESA, 2024). Two polarizations are used in this analysis: single co-polarization (vertical transmit/vertical receive, or VV) and dual-band cross-polarization

(vertical transmit/horizontal receive, or VH). These polarizations are differentially sensitive to different types of scattering mechanisms generated by ground targets, such as double-bounce scattering generated by standing buildings or rough surface scattering generated by rubble (Meyer, 2019; Canada, 2008). Previous studies have found that a combination of VV and VH polarizations improved building damage detection accuracy (Karimzadeh and Mastuoka, 2017).

Ground Range Detected (GRD) scenes acquired in Interferometric Wide-Swath mode are accessed in Google Earth Engine at a spatial resolution of 10 meters per pixel (ESA, 2024). The collection is updated in near real time and undergoes orthorectification, thermal noise removal, and radiometric calibration prior to ingestion into the Earth Engine catalog (ESA, 2024). Speckle filtering is conducted prior to analysis using a Lee Filter, which preserves edges while reducing speckle noise by smoothing sharp intensity variations using local statistics calculated by a sliding kernel (Lee et al., 1994). Because the urban areas in this study are all in relatively flat areas, the terrain corrections applied in the orthorectification process are sufficient. However, if analysis is to be undertaken in mountainous regions, Radiometric Terrain Flattening (RTF) should be applied; a python implementation of the RTF process elaborated in Vollrath et al. (2020) is included as an optional pre-processing step in the PWTT API.

#### 2.1.2. Annotated building footprints

Accuracy assessments are carried out using an original dataset of 904,977 annotated building footprints, spanning 30 cities in five different countries. This dataset was compiled by spatially joining damage annotations from the United Nations Satellite Centre (UNOSAT) with data on building footprints. This allows for the generation of more meaningful accuracy statistics, and a more stringent test of model accuracy than previous studies. Accuracy assessment at the building level follows the standards set by xBD dataset (Gupta et al., 2019).

Building footprint data is sourced from the Microsoft Building Footprints dataset, which consists of over 1 billion building footprints derived from high resolution satellite imagery around the world using deep learning (Microsoft, 2024). For the Middle East and Europe, buildings are detected with a precision of 95% and recall of 85%. A manual review of the footprints found them to be generally high quality, but that many of the very small ( $< 50 \text{ m}^2$ ) building footprints

**Table 1**  
Validation dataset.

| Country   | City            | Footprints | % Damaged | Annotations date |
|-----------|-----------------|------------|-----------|------------------|
| Palestine | Gaza            | 228 728    | 71.91     | 2025-04-04       |
| Ukraine   | Kherson         | 7302       | 0.41      | 2022-10-14       |
| Ukraine   | Antonivka       | 905        | 0.99      | 2022-10-14       |
| Ukraine   | Lysychansk      | 20 246     | 7.42      | 2022-09-21       |
| Ukraine   | Avdiivka        | 7262       | 8.7       | 2022-09-20       |
| Ukraine   | Melitopol       | 32 373     | 0.23      | 2022-08-02       |
| Ukraine   | Sievierodonetsk | 5970       | 24.32     | 2022-07-27       |
| Ukraine   | Kramatorsk      | 21 880     | 0.22      | 2022-07-24       |
| Ukraine   | Mykolaiv        | 60 467     | 0.34      | 2022-07-21       |
| Ukraine   | Rubizhne        | 8899       | 33.67     | 2022-07-09       |
| Ukraine   | Sumy            | 28 265     | 0.08      | 2022-07-08       |
| Ukraine   | Shchastia       | 1293       | 1.93      | 2022-07-07       |
| Ukraine   | Kremenchuk      | 27 497     | 0.11      | 2022-06-29       |
| Ukraine   | Kharkiv         | 107 976    | 0.85      | 2022-06-15       |
| Ukraine   | Mariupol        | 18 446     | 31.42     | 2022-05-12       |
| Ukraine   | Borodyanka      | 3912       | 3.91      | 2022-05-02       |
| Ukraine   | Volnovakha      | 8616       | 12.15     | 2022-04-26       |
| Ukraine   | Moschun         | 603        | 55.39     | 2022-03-31       |
| Ukraine   | Vorzel          | 959        | 1.15      | 2022-03-31       |
| Ukraine   | Irpin           | 7242       | 11.28     | 2022-03-31       |
| Ukraine   | Hostomel        | 4175       | 14.04     | 2022-03-31       |
| Ukraine   | Bucha           | 5739       | 1.6       | 2022-03-31       |
| Ukraine   | Okhtyrka        | 15 905     | 0.4       | 2022-03-25       |
| Ukraine   | Trostanets      | 8913       | 0.72      | 2022-03-25       |
| Ukraine   | Chernihiv       | 29 929     | 3.25      | 2022-03-22       |
| Ukraine   | Makariv         | 3514       | 4.01      | 2022-03-16       |
| Syria     | Raqqa           | 24 689     | 45.36     | 2017-10-21       |
| Iraq      | Mosul           | 137 794    | 11.51     | 2017-08-04       |
| Sudan     | Yei             | 9608       | 1.97      | 2017-03-05       |
| Syria     | Aleppo          | 65 870     | 30.82     | 2016-09-18       |
| All       | All             | 904 977    | 12.67     |                  |

By joining manual damage annotations from UNOSAT with building footprints using a 10 meter tolerance, 930,122 labeled building footprints spread across 30 cities in 5 different conflict zones are used for validation.

appeared to be false positives (cars/trucks, garden sheds, or errant geometries), and were removed. For a discussion of the availability and reliability of global, multi-source open access building footprints, see Fig. A.2 in Appendix A.

A building footprint is labeled as damaged if it intersects with a UNOSAT damage annotation point, and labeled undamaged otherwise. UNOSAT annotations are generated manually on the basis of high resolution optical satellite imagery (Unosat, 2022c,a, 2016, 2017b,a, 2022b,d,e,f,g,h, 2024). These annotations are the most commonly used source of labeled data in studies of battle damage, and are generally regarded to be of high quality (Mueller et al., 2021; Kahraman et al., 2016; Witmer, 2015; Boloorani et al., 2021; Braun, 2018; Huang et al., 2023). However, because the imagery used for annotation is nadir, some damage caused to the sides of buildings may be missed.

The validation dataset was constructed using all available UNOSAT building damage annotations where:

1. The damage was the result of armed conflict, carrying the 'Complex Emergency' tag on the UNOSAT website
2. At least 12 months of pre-war Sentinel-1 imagery is available.
3. Reliable building footprint data is available.

These criteria result in the inclusion of all available UNOSAT annotations for Ukraine, the Gaza Strip, and Iraq. In Sudan, UNOSAT conducted a damage assessment for El-Fasher on 2024-07-31. However, the city was only covered by Sentinel-1B, which stopped transmitting data in December 2021 due to a system failure and could therefore not be included in this sample.

In Syria, damage annotations are available for 8 cities. However, the Syrian civil war started 2011, meaning that Sentinel-1 only began imaging the area nearly four years after the beginning of the war, by which time many cities had already been leveled. This creates a twofold problem; Firstly, most building footprint datasets use imagery that was taken later than 2011, meaning that areas that were destroyed

early in the war lack building footprints in most open-access datasets. The second (and more fundamental) problem, is simply that damage preceding the PWTT reference period is undetectable. Though both Raqqa and Aleppo sustained significant damage prior to the availability of Sentinel-1 imagery, the majority of the damage sustained by both cities occurred in 2017 and 2016, respectively (Unosat, 2016, 2017a). As such, the two cities are included in the main results for added external validity, despite poor quality building footprints and pre-existing damage. Appendix A.3.4 in the Appendix generates PWTT damage estimates using 32-meter grid cells across 6 syrian cities, for a direct comparison to the study by Mueller et al. (2021).

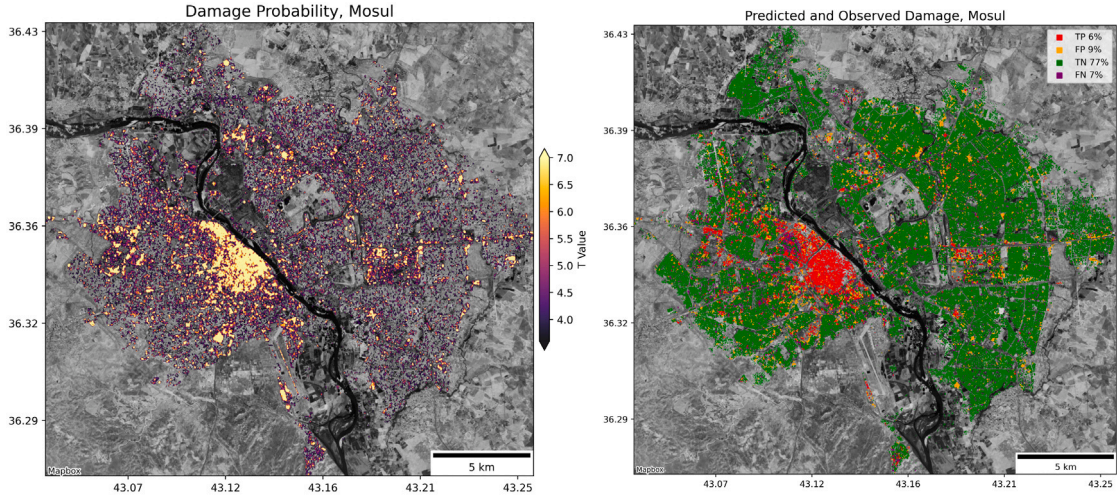
The resulting dataset has substantial coverage across four different conflict zones, with 25% of the footprints located in the Gaza Strip, 48% in Ukraine, 15% in Iraq, 10% in Syria, and 1% in Sudan. These cities encompass a wide range of different climate and atmospheric conditions, building types, urban morphologies, and damage levels, ranging from just 0.08% damage in Sumy to 72% damage in Gaza. Table 1 provides summary statistics for the verification dataset.

While this dataset enables a comprehensive test of geographic generalization, temporal generalization is just as important in the context of long-running wars. To this end, an additional annotated multitemporal building damage dataset was compiled for Gaza, where UNOSAT conducted six assessments between October 2023 and May 2024. This dataset is also used to compare the results of the PWTT against a deep learning approach in Section 3.2.

## 2.2. The Pixel-Wise T-test

Fig. 2 demonstrates the change in backscatter amplitude for a destroyed building in Mariupol, Ukraine, following its destruction. The corresponding Sentinel-1 pixel has a low standard deviation in both the pre-and post-war periods, but experiences a large change in mean amplitude. The T-Test is a simple signal-to-noise ratio that measures the difference between the means of two samples adjusted by the standard





**Fig. 3.** Damage prediction and accuracy assessment, Mosul.

The damage probability raster for Mosul is shown on the left. When joined with the labeled building footprints (right), True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. Percentages in the legend reflect the proportion of building area in each category. Though Mosul has one of the lowest F1 score (0.42) out of the entire sample, the PWTT still correctly localizes damage in the city. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

deviation within each sample. The integration of multi-temporal pixel standard deviation into the test allows for better isolation of damage, rather than general anthropogenic change as detected by bi-temporal algorithms (Canty, 2019); Airports and train stations are likely to have high pixel variance over time, meaning that these areas are unlikely to achieve high T-values. As such the T-test is well suited to the detection of building damage resulting from conflict.

The first step in the calculation of the test statistic is the selection of a pre-war reference period ( $\tau = 0$ ) and a post war inference period ( $\tau = 1$ ) using a cutoff date (e.g. the beginning of hostilities in a given area). Fig. A.7 in the Appendix shows the selection of reference and inference periods for the cities in the sample along with the weekly number of clashes therein. In all cases, the start of the inference period corresponds to the date of the UNOSAT damage assessment used for validation, and spans one month thereafter. The reference periods for Gaza and all Ukrainian cities span one year prior to the onset of hostilities (2024-10-10 and 2022-02-24, respectively). For Mosul, Raqqa, and Aleppo, the reference periods span the first year Sentinel-1 imagery was available (2014-10-10 to 2015-10-10).

The pre-war reference period spans a full year (~120 images) in order to capture the effect of seasonal changes in the dielectric properties of the target (e.g., the presence of snow). The inference period uses one months' worth of imagery (~10 images)– shorter time periods result in too small a sample for inference, and the use of longer time periods increases the risk of damage accruing during the inference period. For rapid response scenarios, a Pixel-Wise Z-Score can be computed using the first available post-event Sentinel-1 image, by subtracting the pre-war mean from the post-event image, and dividing by the pre-war standard deviation. Results from this specification are reported in Table 4. For optimal performance, there should be minimal change during both sample periods. This is the case for Ukraine and Gaza which both featured an abrupt onset of hostilities. However, the algorithm still performs well when there is substantial damage occurring during the reference period, as is the case for the cities in Syria and Iraq.

Following the selection of a pre-conflict reference period and a post-conflict inference period, the calculation of pixel-level t-values proceeds as follows. To ensure consistency in the look angle, Sentinel-1 scenes are disaggregated using the Relative Orbit Number metadata property, which indicates scenes with the same footprint and incidence angle. The mean backscatter amplitude  $\bar{x}$  for each pixel  $x$  is calculated for all

unique combinations of orbital trajectory  $\omega$ , polarization  $\pi$ , and time period  $\tau$ . Variable  $n$  denotes the number of Sentinel-1 scenes.

$$\bar{x}_{\omega\pi\tau} = \frac{1}{n_{\omega\pi\tau}} \sum_{i=1}^{n_{\omega\pi\tau}} x_{\omega\pi\tau} \quad (1)$$

The corresponding pixel-level standard deviation  $s$  is also calculated.

$$s_{\omega\pi\tau} = \sqrt{\frac{1}{n_{\omega\pi\tau} - 1} \sum_{i=1}^{n_{\omega\pi\tau}} (x_{\omega\pi\tau} - \bar{x}_{\omega\pi\tau})^2} \quad (2)$$

A standard two-tailed T-test is then computed for each combination of orbital trajectory (at least one ascending and one descending, though more are sometimes available) and polarization (VV/VH).

$$t_{\omega\pi} = \frac{\bar{x}_{\omega\pi\tau_0} - \bar{x}_{\omega\pi\tau_1}}{\sqrt{\frac{s_{\omega\pi\tau_0}^2}{n_{\omega\pi\tau_0}} + \frac{s_{\omega\pi\tau_1}^2}{n_{\omega\pi\tau_1}}}} \quad (3)$$

There are thus at least four t-values per pixel, depending on the number of scenes with distinct orbital trajectories that overlap with the area of interest. The final change statistic  $T$  is the maximum of the absolute t-values, because the directionality of the change in backscatter resulting from the destruction of a building is unknown a-priori, and because damage may only be visible to one side of the building.

$$T = \max\{|t_{\omega\pi}|\} \quad (4)$$

Finally, the raw output is smoothed using convolutional kernels with a radius of 50, 100, and 150 m, and the mean of all four layers is calculated. This enables better delineation of damage, as buildings on the periphery of a damaged area will have lower values in the convolved layers, and reduces noise. The final result is a damage probability raster, as shown in Fig. 3. A process diagram for the PWTT is available in Fig. A.1 in Appendix A.

Assuming that damage generally increases surface roughness, depending on a building's geometry and orientation relative to the satellite, the transition from a standing structure to rubble could either lead to an increase or decrease in backscatter amplitude. For example, large structures with flat roofs such as warehouses or shopping centers might experience an increase in backscatter amplitude after being damaged as the scattering mechanism transitions from specular reflection to diffuse

**Table 2**  
Building-level damage detection accuracy.

| Country   | Location        | AUC   | Accuracy | F1    | Precision | Recall | N       |
|-----------|-----------------|-------|----------|-------|-----------|--------|---------|
| Palestine | Gaza            | 78.06 | 74.95    | 82.07 | 74.09     | 91.98  | 228 728 |
| Ukraine   | Kherson         | 89.52 | 84.46    | 14.39 | 8.78      | 39.73  | 7302    |
| Ukraine   | Antonivka       | 42.3  | 89.41    | 2.83  | 2.15      | 4.14   | 905     |
| Ukraine   | Lysychansk      | 75.92 | 79.1     | 47.42 | 40.44     | 57.32  | 20 246  |
| Ukraine   | Avdiivka        | 65.89 | 60.56    | 38.25 | 27.22     | 64.32  | 7262    |
| Ukraine   | Melitopol       | 65.47 | 96.44    | 2.31  | 1.49      | 5.05   | 32 373  |
| Ukraine   | Sievierodonetsk | 70.66 | 55.23    | 63.0  | 48.04     | 91.49  | 5970    |
| Ukraine   | Kramatorsk      | 87.01 | 97.22    | 8.08  | 12.77     | 5.91   | 21 880  |
| Ukraine   | Mykolaiv        | 82.27 | 98.14    | 23.23 | 19.48     | 28.76  | 60 467  |
| Ukraine   | Rubizhne        | 78.53 | 73.78    | 68.06 | 65.85     | 70.43  | 8899    |
| Ukraine   | Sumy            | 66.05 | 99.5     | 4.22  | 4.29      | 4.16   | 28 265  |
| Ukraine   | Shchastia       | 72.56 | 97.41    | 40.64 | 74.93     | 27.88  | 1293    |
| Ukraine   | Kremenchuk      | 92.8  | 98.92    | 3.2   | 3.44      | 2.99   | 27 497  |
| Ukraine   | Kharkiv         | 80.3  | 96.51    | 27.04 | 23.41     | 32.0   | 107 976 |
| Ukraine   | Mariupol        | 71.72 | 65.34    | 69.7  | 60.57     | 82.06  | 18 446  |
| Ukraine   | Borodyanka      | 88.21 | 93.96    | 53.92 | 47.31     | 62.68  | 3912    |
| Ukraine   | Volnovakha      | 75.03 | 78.63    | 46.09 | 39.6      | 55.12  | 8616    |
| Ukraine   | Bucha           | 90.43 | 98.2     | 72.18 | 98.04     | 57.11  | 5739    |
| Ukraine   | Hostomel        | 84.24 | 77.51    | 67.53 | 57.56     | 81.69  | 4175    |
| Ukraine   | Vorzel          | 83.1  | 96.39    | 46.84 | 47.38     | 46.3   | 959     |
| Ukraine   | Moschun         | 72.47 | 65.44    | 75.14 | 63.36     | 92.3   | 603     |
| Ukraine   | Irpin           | 84.81 | 83.26    | 59.43 | 53.32     | 67.13  | 7242    |
| Ukraine   | Trostanets      | 85.56 | 98.51    | 69.17 | 81.78     | 59.93  | 8913    |
| Ukraine   | Okhtryka        | 89.06 | 98.87    | 53.92 | 71.6      | 43.24  | 15 905  |
| Ukraine   | Chernihiv       | 84.62 | 93.98    | 45.1  | 48.52     | 42.13  | 29 929  |
| Ukraine   | Makariv         | 82.6  | 90.7     | 39.55 | 35.28     | 45.0   | 3514    |
| Syria     | Raqqa           | 76.35 | 68.31    | 75.01 | 65.79     | 87.22  | 24 689  |
| Syria     | Aleppo          | 74.26 | 64.92    | 64.18 | 52.48     | 82.59  | 65 870  |
| Iraq      | Mosul           | 75.97 | 82.98    | 42.26 | 38.47     | 46.89  | 137 794 |
| Sudan     | Yei             | 79.14 | 95.75    | 15.49 | 11.38     | 24.24  | 9608    |
| All       | All             | 87.07 | 80.78    | 64.71 | 59.2      | 71.36  | 904 977 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. For each location, a threshold value of  $T$  is selected using a precision–recall curve to generate a binary damage classification. All statistics are weighted by building area.

scattering (Huang et al., 2023). Conversely, a free-standing house that previously generated double-bounce scattering would experience a drop in backscatter amplitude after becoming destroyed, as is likely the case in Fig. 2.

### 2.3. Damage classification

After the generation of the damage probability raster, building-level inference is carried out by calculating the mean pixel value within each building footprint, and setting a threshold value for binary damage classification (damaged/undamaged).

A key advantage of the PWTT is that the damage probability raster contains non-arbitrary values. A threshold value of  $T$  can be chosen according to the desired statistical significance (e.g.  $T > 2.7$  where  $n = 40$  at the 99% confidence level). This approach classifies statistically significant changes that occurred during the post-conflict period as damage. T-values also generalize across domains, since the T-test is a measure of relative change and accounts for absolute differences in backscatter values across geographic areas. The threshold value of  $T$  can also be selected empirically, using a precision–recall curve (as shown in Fig. 4), allowing for the selection of a more context-aware threshold for a given geography.

## 3. Results

### 3.1. Building-level damage

Building level accuracy assessment is carried out as follows. Given a building footprint  $F$ , UNOSAT damage annotation  $D$ , and PWTT change threshold  $\tau$ , a building is labeled damaged if it intersects with a damage label ( $F \cap D \neq \emptyset$ ), and is predicted to be damaged if the average

value of  $T$  within the building footprint exceeds the damage threshold ( $\bar{T}(F) > \tau$ ), therefore:

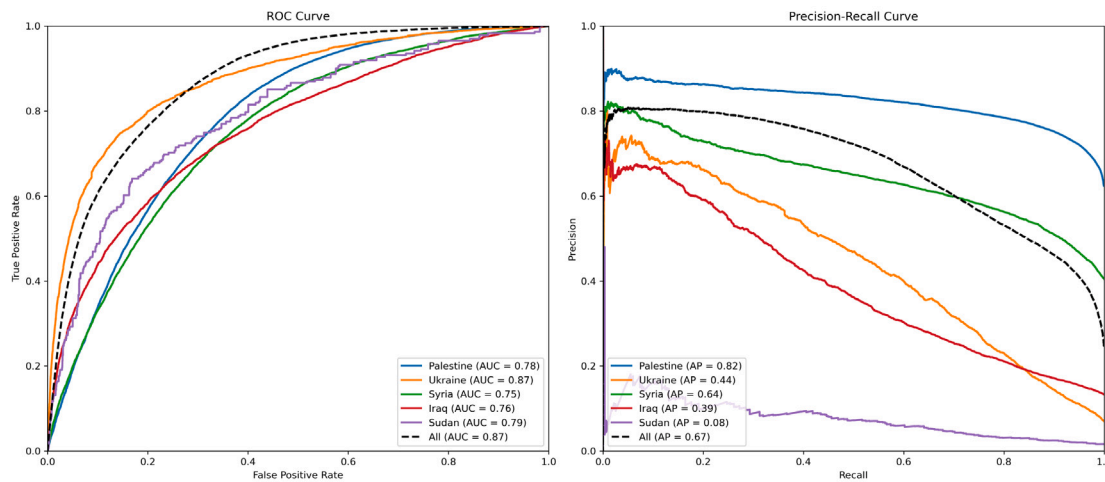
$$\begin{cases} \text{True Positive (TP):} & F \cap D \neq \emptyset \quad \text{and} \quad \bar{T}(F) > \tau, \\ \text{True Negative (TN):} & F \cap D = \emptyset \quad \text{and} \quad \bar{T}(F) \leq \tau, \\ \text{False Positive (FP):} & F \cap D = \emptyset \quad \text{and} \quad \bar{T}(F) > \tau, \\ \text{False Negative (FN):} & F \cap D \neq \emptyset \quad \text{and} \quad \bar{T}(F) \leq \tau. \end{cases}$$

Fig. 3 demonstrates the process of accuracy assessment in Mosul. The damage probability raster on the left predicts heavy damage to the Old City on the south bank of the Tigris River. The plot on the right joins the damage predictions with the labeled building footprints, showing True Positives in red, True Negatives in green, False Positives in orange, and False Negatives in purple. The PWTT accurately localizes damage, not just in the Old City but in isolated pockets to the West as well. This process is carried out for all 30 cities, and building-level accuracy metrics are reported in Table 2.

Accuracy – the ratio of correct predictions to the total – is the simplest gauge of model performance, but is strongly affected by class imbalance; in a city where only 1% of buildings are damaged, a model blindly predicting everything to be undamaged would still garner 99% accuracy. By ignoring true negatives, the F1 score is more robust to class imbalance and is a more widely reported accuracy statistic in the study of building damage. It consists of the harmonic mean of Precision (the proportion of positive predictions that are correct) and Recall (the proportion of positive labels that are retrieved).

Using a single threshold value of  $T$  ( $> 3.2$ ) for all of the building footprints in the sample, the PWTT achieves an F1 score of 64.7. The highest F1 scores are achieved in Gaza (82%) and Raqqa (75.2%), while the lowest are observed in Antonivka (2.8%) and Melitopol (2.3%).

Though less biased than simple accuracy, the F1 score is still sensitive to class imbalance. When comparing these results to the percentage



**Fig. 4.** Country-level ROC and Precision–Recall curves.

The Receiver Operating Characteristic (ROC) curve plots the true positive and false positive rates at each threshold value of  $T$ . The Precision–Recall curves does the same with precision and recall.

of damaged buildings in a given city, it becomes apparent that F1 scores are positively correlated with the extent of destruction in a city: 72% of buildings in Gaza are damaged, compared with less than 0.3% of buildings in Melitopol. As such, comparing F1 scores across cities can be misleading: given a classifier with a non-zero false positive rate, F1 approaches zero with the addition of new observations of the negative class. When a balanced sample (with the same number of damaged and undamaged buildings) is used, the F1 score for the full dataset increases to 80% (see Appendix, Table A.8).

The Area Under the Curve is insensitive to changes in the class distribution, and therefore provides a more comparable measure of discrimination across contexts. It is generated by calculating the true positive rate and the false positive rate at all possible threshold values of  $T$  and plotting the resulting Receiver Operating Characteristic (ROC) curve, shown in Fig. 4. AUC of 0.5 suggests no discrimination, AUC of 1 indicates perfect discrimination, and in the field of Oncology, an AUC of “0.8 to 0.9 is considered excellent” (Mandrekar, 2010). The AUC across all of Ukraine is 0.87, and 0.78 for the Gaza Strip. AUC is lower for cities in Syria and Iraq, likely due to the fact that the reference periods used in the generation of the PWTT contained substantial fighting (and therefore, destruction), resulting in noisier estimates; Sentinel-1 imagery only became available over a year after the onset of hostilities in Iraq, and three years after the onset of the Syrian civil war.

Despite having the lowest AUC scores in the sample, the results for Syria rival deep learning approaches to conflict damage detection in out-of-sample scenarios. In their analysis of damage to six Syrian cities during the civil war, Mueller et al. (2021) employ 50 cm resolution optical imagery, a convolutional neural network, and a second stage random forest model. They achieve an average AUC of 0.87 across all cities. But when a model trained only on Aleppo is deployed on other cities, the average AUC drops to 0.7. Direct comparison using the data from Mueller et al. (2021) was not possible due to the inaccessibility of the high resolution satellite imagery used in the study. A detailed comparison of the PWTT with the results reported by Mueller et al. (2021) using the full sample of six Syrian cities is reported in Appendix A.3.4 in Appendix A. Though the deep learning approach is more accurate when trained and tested on the same city, the PWTT outperforms the deep learning approach in two out of the four cities in which the model was trained in a different city, despite being unsupervised and using lower resolution open-access satellite imagery. However, these results are approximate because the source data for the Mueller study was unavailable, and there are some discrepancies between the samples in both studies. The following section conducts a one-to-one comparison of the PWTT with the current state of the art model in building damage detection using deep learning.

### 3.2. Temporal generalization and comparison to deep learning

The results in the previous subsection demonstrate that the PWTT is capable of strong geographic generalization that can match the performance of deep learning methods at analogous tasks. But in the context of long-running wars, the temporal dimension of generalization is just as important as the spatial, unlike a natural disaster where a shock is located at one point in time.

This subsection systematically compares the PWTT’s ability to generalize not only geographically but temporally, against the state of the art ChangeOS deep learning model deployed on high resolution satellite imagery. Inference is carried out on the Gaza Strip and validated using over 1.3 million labeled building footprints drawn from UNOSAT assessments carried out at six different time intervals between October 2023 and May 2024. All PWTT results use the same reference period, spanning one year prior to October 10th, 2023, and an inference period of one month beginning on the date of the UNOSAT assessment.

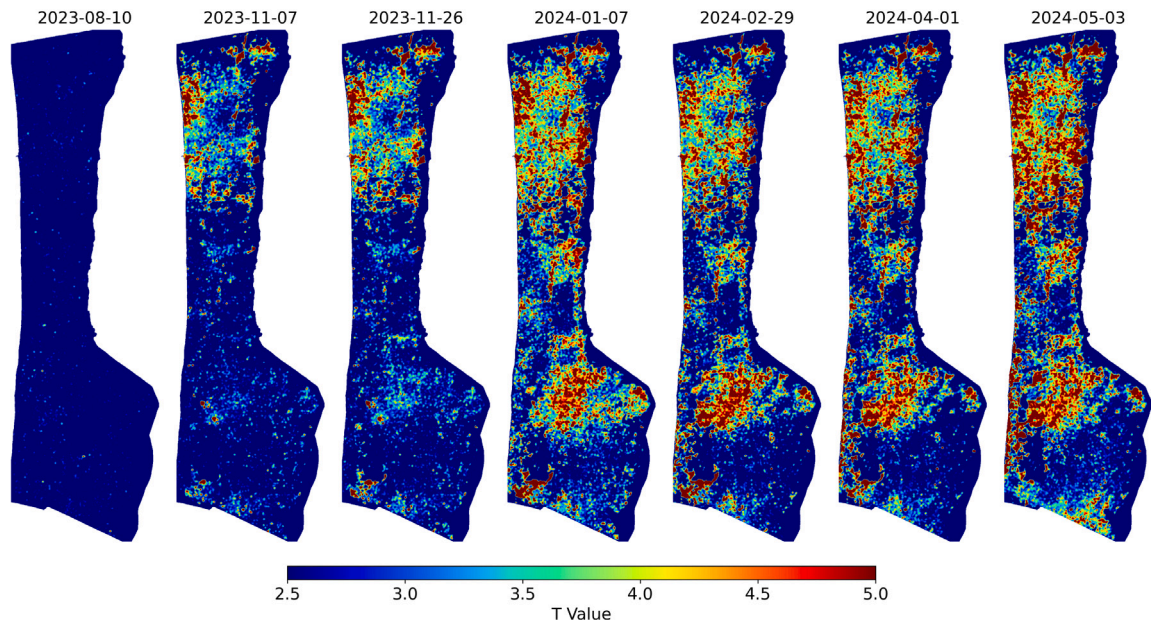
Fig. 5 displays six damage probability rasters calculated on the dates of the UNOSAT assessments, as well as one placebo test carried out two months before the onset of hostilities.

The leftmost plot shows virtually no statistically significant changes during the placebo period. The two PWTT results for November 2023 show high damage probability across swathes of northern Gaza, as the southern city of Khan Younis had originally been designated a safe zone by the Israeli Defense Forces (IDF). By the end of the month, most of the health and education facilities in Gaza had been destroyed (Asi et al., 2024). On December 1st, 2023 the IDF announced a ground invasion of Khan Younis (Wintour et al., 2023); the subsequent damage probability raster shows large scale damage across the southern city. The damage probability rasters faithfully reproduce the spatial dynamics of the conflict over time.

The PWTT results are compared to ChangeOS, a deep object-based semantic change detection framework developed by Zheng et al. (2021). Beyond achieving state of the art performance for building damage assessment from natural disasters on the xBD dataset, the authors emphasize that the ChangeOS framework “has a superior generalization ability for man-made disasters.” They demonstrate this by using a model pre-trained on the xBD dataset to conduct damage assessment following the 2020 Beirut explosion and the Bata Military barracks explosion, achieving better performance than both a UNet+ResNet and Siamese-UNet architectures.

To conduct damage assessment using deep learning, high resolution (50 cm/px) Planet SkySat images were accessed, enabling the creation





**Fig. 5.** Damage probability over time, Gaza.

The Pixelwise T-Test is calculated at 7 time intervals across the Gaza Strip using a 1 month inference period from the date indicated above each raster. The leftmost plot is a placebo test, while each subsequent plot is calculated on the date of a UNOSAT damage assessment. The colormap corresponds to T-values which proxy for damage. Based on the validation data, precision in light blue areas stands at 79%, and rises to 86% in dark red areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 3**  
Gaza strip multitemporal building-level damage detection accuracy.

| Date       | ChangeOS     |           |        |       | PWTT         |           |        |       |
|------------|--------------|-----------|--------|-------|--------------|-----------|--------|-------|
|            | F1           | Precision | Recall | AUC   | F1           | Precision | Recall | AUC   |
| 2023-11-07 | <b>22.52</b> | 13.17     | 77.70  | 77.73 | <b>46.67</b> | 37.40     | 62.03  | 83.34 |
| 2023-11-26 | <b>32.30</b> | 20.26     | 79.63  | 77.14 | <b>54.55</b> | 45.30     | 68.54  | 83.70 |
| 2024-01-07 | <b>50.33</b> | 36.58     | 80.66  | 75.11 | <b>64.25</b> | 53.10     | 81.33  | 81.33 |
| 2024-02-29 | <b>63.99</b> | 53.27     | 80.13  | 77.86 | <b>75.42</b> | 75.24     | 75.59  | 84.18 |
| 2024-04-01 | <b>63.56</b> | 53.08     | 79.19  | 77.53 | <b>75.91</b> | 74.26     | 77.64  | 84.24 |
| 2024-05-03 | <b>67.09</b> | 57.93     | 79.71  | 75.92 | <b>77.75</b> | 73.82     | 82.13  | 82.69 |
| Average    | <b>49.97</b> | 39.05     | 79.50  | 76.88 | <b>65.76</b> | 59.86     | 74.54  | 83.25 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. For the PWTT, binary damage classification is conducted using a threshold T-value of 2.9. For ChangeOS, the three categories of damage output by the model were collapsed into a single category. AUC is generated using T-values for the PWTT, and the 0-4 damage scale for ChangeOS. The PWTT results outperform ChangeOS for all time periods, across all measures.

of seven mosaics covering the Gaza Strip: one prior to the conflict, and six assembled during the month corresponding to each UNOSAT damage assessment. A  $500 \times 500$  meter grid was generated across Gaza to generate pre- and post-war patches of high-resolution optical imagery. Using the pixel mean and standard deviation from the fully clear pre-war mosaic, the best available post-war patch was selected by minimizing average pixel-level z-score, thereby avoiding cloudy and hazy scenes where possible. Following patch generation, inference was conducted using the pre-trained ChangeOS model with the ResNet-101 backbone. ChangeOS outputs 4 classes (no damage, minor damage, major damage, and destroyed), which were re-categorized into damage/no-damage.

These damage predictions from the PWTT and ChangeOS compared to a validation dataset of 1,321,038 labeled building footprints across Gaza at 6 different time periods, and the results are presented in Table 3.

The results in Table 3 demonstrate that the PWTT outperforms the state-of-the-art deep learning model, ChangeOS, in all time periods under observation, on all accuracy metrics. F1 tends to increase as the number of damage buildings (and therefore class balance) increases. AUC for both the PWTT and ChangeOS remain highly stable over time, suggesting consistent discriminant ability over time. Across all five time

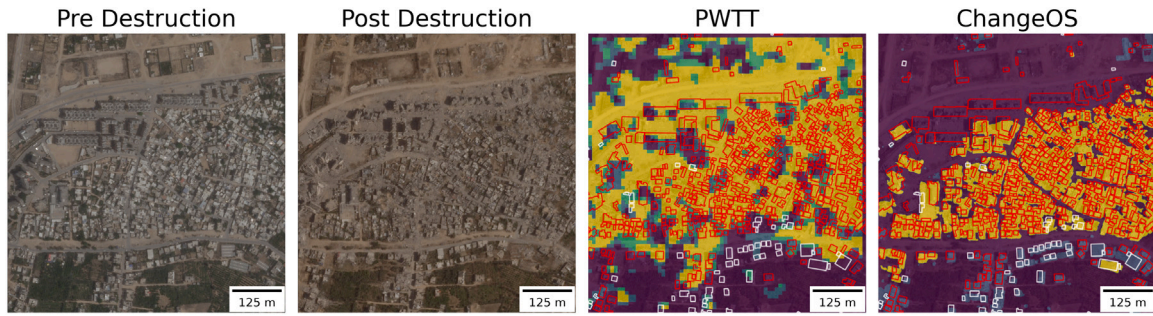
periods, the PWTT's average F1 and AUC are 15% and 7% higher than ChangeOS, respectively.

There are several possible reasons for the degradation in performance experienced by ChangeOS, which are not experienced by the PWTT. Clouds are a perennial problem for optical imagery, and wars can make their own weather; Pyrocumulus clouds generated by fires in urban areas are characterized by thick black smoke that can obscure entire swathes of a city. Similarly, airborne dust and haze from heavy bombing can blanket wide areas. Despite being able to select the highest quality patch from multiple post-war images, many of these occlusions were unavoidable. In practice, a single post-war image may be available.

On the other hand, a significant limitation of the PWTT relative to most deep learning models including ChangeOS is that the latter only requires a single post-event image while the PWTT requires at least one month of imagery for maximum accuracy. To test the performance of the PWTT under rapid-response conditions, Table 4 reports the results of damage detection carried out using a single post-event image. Table A.7 in the appendix reports city-level results for this specification.

In this case, the same one-year reference period is used to calculate a Pixel-Wise Z-Score (PWZS) by subtracting the pre-war mean from the post-event image, and dividing by the pre-war standard deviation.





**Fig. 6.** Comparison of PWTT and ChangeOS damage detection in Izbet Beit Hanoun, Gaza.

The results of the Pixel-Wise T-Test are compared to the ChangeOS damage detection model. On the right two subplots, buildings labeled as damaged by UNOSAT are overlaid in red, while undamaged buildings are shown in white. Predicted damage is shown in yellow. Though it has a higher resolution, ChangeOS fails to localize irregularly shaped buildings, such as the rows of apartment complexes to the north of the scene. Pre and Post-conflict images ©2024 Planet Labs PBC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 4**  
Country-level results for modified PWTT architectures.

| Country   | 1 Post-event image |       |           |        | 1 Month, gradient boosting |       |           |        |         |
|-----------|--------------------|-------|-----------|--------|----------------------------|-------|-----------|--------|---------|
|           | AUC                | F1    | Precision | Recall | AUC                        | F1    | Precision | Recall | N       |
| Palestine | 75.81              | 80.77 | 71.75     | 92.38  | 78.06                      | 87.93 | 86.69     | 89.20  | 228,728 |
| Sudan     | 69.72              | 10.95 | 6.63      | 31.45  | 79.14                      | 30.40 | 58.46     | 21.32  | 9,608   |
| Ukraine   | 80.35              | 39.21 | 38.84     | 39.59  | 87.30                      | 55.43 | 67.60     | 47.03  | 415,991 |
| Syria     | 64.33              | 55.89 | 42.09     | 83.17  | 75.10                      | 70.18 | 71.80     | 68.64  | 90,559  |
| Iraq      | 70.53              | 35.64 | 30.12     | 43.65  | 75.97                      | 49.27 | 74.55     | 36.80  | 137,794 |
| All       | 81.13              | 57.58 | 49.58     | 68.66  | 87.07                      | 75.24 | 78.69     | 72.08  | 904,977 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. Labels for Gaza are taken from the 2025-04-04 UNOSAT assessment. All statistics are weighted by building area. For the Pixel-Wise Z-Scores, a single cutoff value of Z is employed per country. For the gradient boosting model, the 80–20 train-test split is drawn from building footprints across all cities in a given country.

Though F1 and AUC are each reduced by roughly 6% relative to the 1-month T-Test, results still indicate a robust ability to detect damage buildings: in Gaza, even using a Z-score calculated from a single post-event image outperforms ChangeOS by 3% in terms of AUC and 7% in terms of the F1 score.

The underperformance of ChangeOS in Gaza likely relates to urban morphology. Generalization remains a persistent challenge for deep learning architectures in this domain. In their own analysis of the 2020 Beirut explosion, [Zheng et al. \(2021\)](#) find that model performance degrades significantly because “the large off-nadir angle makes the tall urban buildings appear more inclined in the image, thereby causing a more serious misregistration problem.” This is also the case in Gaza, which is an extremely dense urban environment, and the available high-resolution imagery comes in a variety of look angles. Furthermore, as a conflict wears on, the angle of the sun and the shadows cast by the buildings also changes substantially, potentially leading to error.

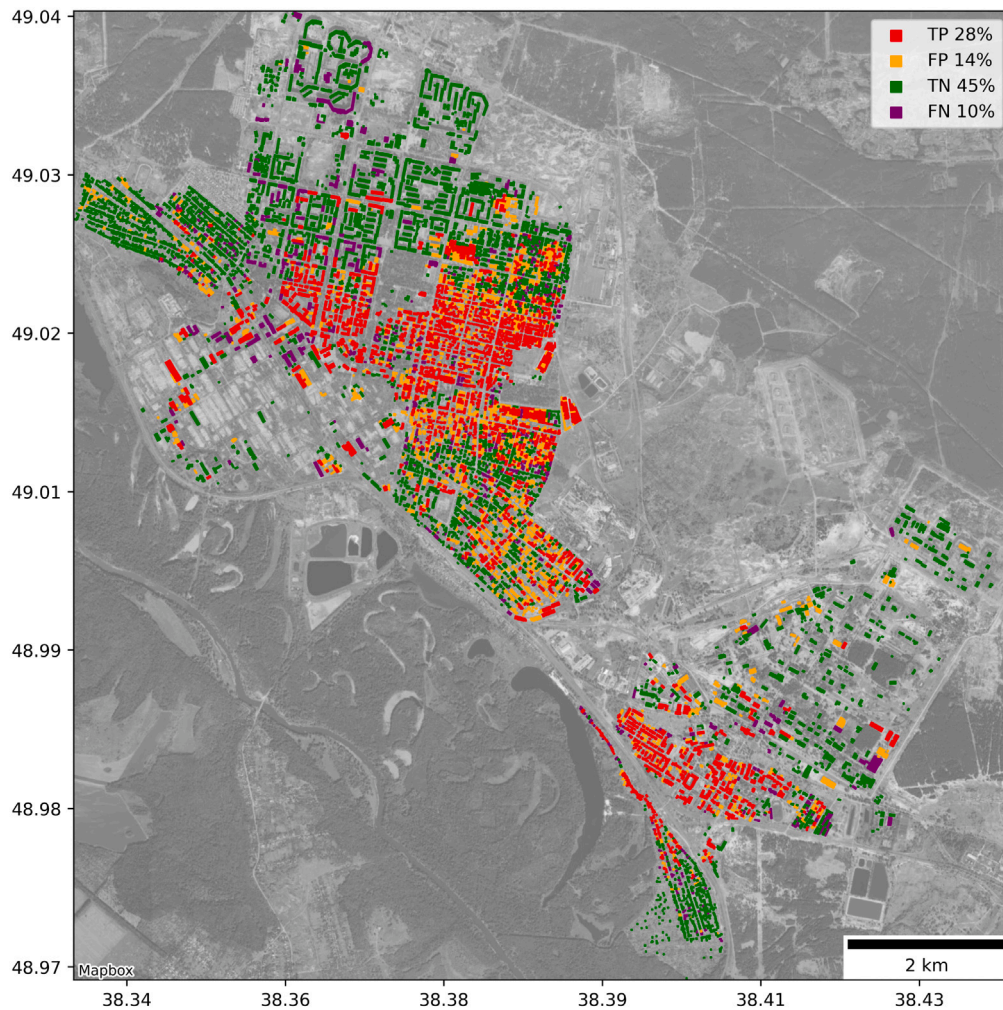
While one advantage of ChangeOS is that it does not require external building footprint data and can produce building-level damage estimates from imagery alone, the building footprint extraction component of the model also performs poorly out of sample. ChangeOS struggles most in high-density regions of Gaza, and particularly with irregularly-shaped buildings such as apartment complexes, which the model often fails to localize completely as shown in [Fig. 6](#). This is a particularly serious problem, given that the model performs worst in areas with the most affected people.

Some of these problems could potentially be alleviated via a more diverse training regimen; ChangeOS was pre-trained on natural disasters, yet the authors explicitly highlight the model’s ability to generalize to man-made disasters. Fine-tuning in conflict settings may improve quantitative measures of accuracy, but largely defeats the purpose of damage assessment using remote sensing: if damage labels for a city already exist, why use a model to detect damage? In the vast majority of cases, building damage detection via remote sensing is to be undertaken

precisely in areas in which no damage labels exist; in Ukraine for example, a deep learning model requiring fine-tuning in order to function adequately could only produce damage estimates for the handful of cities in which the U.N. has already carried out damage assessments, yet hundreds of cities and towns across the country have been leveled over the course of the war. Thus, field performance requires a model to function consistently out-of-the-box without fine-tuning, as is the case with the PWTT; to quote the authors of the ChangeOS paper, “In the real world, sudden onset major disasters have diverse categories and are unpredictable, which means that a robust building damage assessment model should have a generalization ability for the out-of-distribution disasters that can happen anywhere. Local-scale man-made disasters meet this requirement.” ([Zheng et al., 2021](#), p.4).

Nevertheless, there may be scenarios in which damage labels for a particular city already exist. In this case, it makes sense to use all available information to maximize accuracy. The PWTT is highly extensible, and rather than simply setting a cutoff value of  $T$  for damage classification, lightweight machine learning can be applied to improve performance where damage labels are available. [Table 4](#) reports the results of a gradient boosting model applied to building footprints, each comprising 6 features: the maximum  $T$  value layer, the 50, 100, and 150 meter convolutions thereof, the building’s area, and a 500 m grid-cell dummy. Using an 80–20 train-test split and 10 fold cross validation, the gradient boosting model is able to greatly increase accuracy. In Gaza, the F1 score rises from 82% to 88%. Across all 904,977 building footprints, the F1 score increases from 64% (using a single cutoff value of  $T$ ) to 75% using gradient boosting. This is driven by a reduction in false positives: precision increases from 59% to 78% when gradient boosting is used. The following section explores the nature of false positives from the PWTT in detail.

Using a validation dataset of over 1.3 million buildings taken from six different time periods across the Gaza Strip, the PWTT outperforms the state of the art ChangeOS building damage detection model, even



**Fig. 7.** Predicted and observed damage, Rubizhne, Ukraine.

Predicted damage is compared to observed damage in Rubizhne, Ukraine. True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. Percentages in the legend reflect the proportion of building area in each category. False positives are largely co-located with true positives. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

when a single post-event image is used. The PWTT is far less sensitive to atmospheric occlusions, and the generation of a separate test statistic for each relative orbit number ensures consistency in the look angle and thereby avoids the misregistration problem. Unlike deep learning models which are trained on one geographic context and often deployed in another, the PWTT uses a one year pre-war sample period enabling robust performance across a wide range of geographies. Finally, consistently high AUC values across six time periods suggests that the PWTT maintains strong out-of-the-box performance and consistent discriminant ability over time.

### 3.3. Damage spillovers

Though the PWTT enables robust discrimination between damaged and undamaged buildings, precision is consistently lower than recall for most cities. This suggests that false positives are more common than false negatives, even when selecting an optimal threshold value for damage classification. An analysis of false positives suggests that the majority thereof are driven by spillover effects of actual damage. If building footprints that are neighboring (within 10 meters of) a damaged building are re-classified as being potentially damaged, the F1 score for building damage detection across all 30 cities rises from 64% to 68%.

**Fig. 7** plots predicted versus observed damage at the building level in Rubizhne, Ukraine. The vast majority of false positives (shown in orange) are in close proximity to true positives (shown in red). Across all cities, the median distance between a false positive and the nearest damaged building is just 28 m, and 29% of false positives are direct neighbors of a damaged building (within 10 meters of a damage label). In contrast, the median distance between a true negative and the nearest positive label is over fifteen times greater, at 434 m. The difference is highly statistically significant ( $p < 0.0001$ ). **Fig. A.8** in the Appendix plots the histogram of distances between false positive and true negative predictions to the nearest damaged building.

There are a number of possible reasons for these spillovers. Given that roughly a third of false positives are less than one Sentinel-1 pixel away from a damaged building, spillovers are likely partially driven by resolution-related limitations in the process of precisely delineating between damaged and undamaged buildings. There may also be actual changes on the ground in the areas surrounding a damaged building such as rubble dispersed by the destructive event, or fires impacting nearby vegetation. It is also likely that buildings neighboring those that have been destroyed have actually sustained some lateral damage that was missed by the manual damage annotation process but picked up by the side-looking SAR sensor. Thus, the PWTT correctly localizes

**Table 5**  
Regression results between count of damaged buildings and mean T-value, 500 m Grid-Level.

|                | PWTT                |                      |                     |                      |                      |                     | PWZS                |
|----------------|---------------------|----------------------|---------------------|----------------------|----------------------|---------------------|---------------------|
|                | Palestine           | Ukraine              | Syria               | Iraq                 | Sudan                | All                 | All                 |
| Intercept      | 2.724***<br>(0.010) | −0.840***<br>(0.025) | 1.018***<br>(0.021) | −1.168***<br>(0.032) | −2.255***<br>(0.141) | 0.928***<br>(0.010) | 1.762***<br>(0.008) |
| Mean T-Value   | 0.244***<br>(0.002) | 0.959***<br>(0.009)  | 0.471***<br>(0.006) | 0.741***<br>(0.006)  | 1.165***<br>(0.066)  | 0.356***<br>(0.002) | 0.195***<br>(0.002) |
| Building Count | 0.004***<br>(0.000) | 0.010***<br>(0.000)  | 0.006***<br>(0.000) | 0.007***<br>(0.000)  | 0.008**<br>(0.003)   | 0.004***<br>(0.000) | 0.005***<br>(0.000) |
| Pseudo $R^2$   | 0.72                | 0.35                 | 0.62                | 0.53                 | 0.17                 | 0.75                | 0.71                |
| N              | 1777                | 10391                | 1159                | 1124                 | 1374                 | 15825               | 15825               |

\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ . Country-level regression results between count of damaged buildings in a 500 m grid cell are reported above. The rightmost column uses all available grid-cells, and includes a city-level fixed effect. The heading PWTT indicates results derived from the Pixel-Wise T-Test, while PWZS refers to the Pixel-Wise Z-Score, computed with the first available post-conflict image.

damage within a city, as the majority of false positives generated by the PWTT algorithm result from a slight ( $> 30$  meter) over-prediction of the spatial extent of genuinely damaged areas. This is less of a threat to model utility than if the model falsely predicted entire swathes of a city to be damaged. Where damage labels for a particular city are available, gradient boosting can be used to more precisely delineate damage and greatly reduce false positives, as demonstrated in Table 4.

Beyond the damage spillovers, there are several reasons for which perfect agreement between the damage predictions and the labeled building footprints cannot be achieved. The first set of challenges involves the labeled data itself. False negatives could result from a degree of human error in the process of labeling damaged buildings, particularly in regard to lateral damage that might not be visible from a nadir view. The quality of the Microsoft Buildings dataset, while generally high, is also imperfect; the boundaries of buildings with irregular shapes are sometimes misspecified, and a large number of small geometries that are not buildings have been generated. Many of these were removed in the process of data cleaning, but some remain.

Further false negatives may be attributable to temporal mismatch between the prediction window and the date of the manual labeling; UNOSAT damage labeling is mostly conducted on the basis of a single high resolution satellite image, but the PWTT algorithm utilizes a post-conflict time window of one month following the date of the UNOSAT assessment. Damage occurring during this time window would register in the damage predictions, but would not be recorded in the UNOSAT data.

Finally, the resolution of the satellite imagery used in this analysis is a limiting factor when it comes to the identification of damage to small buildings in particular. Indeed, 31% of the buildings in this sample are smaller than a single Sentinel-1 pixel, and 97% are smaller than four pixels (the resolution at which Interferometric Coherence Change Detection is conducted). Despite these constraints, the PWTT manages to generate building damage assessments that rivals the accuracy of deep learning approaches.

### 3.4. Damage intensity

While the identification of individual damaged buildings is an important output for practitioners, zonal damage assessments are also common; UNOSAT conducts Rapid Damage Assessments in which a 500m-by-500 m grid is drawn over a city, and a grid-cell is labeled as damaged if there is at least one damaged building therein Unosat (2022f). The PWTT performs even better at this task; the binary classification of 500 m grid-cells containing at least one damaged building yields an F1 accuracy of 0.65 using a single cutoff value of  $T$  across all cities, with an overall AUC of 0.84. Table A.9 in the Appendix reports city-level accuracy statistics.

However, with such large cells there is a high likelihood of observing multiple damaged buildings within the same cell; while 55% of the grid cells in the global sample contain zero damaged buildings, some grid cells contain up to 823 damaged buildings. Fig. 8 shows the

spatial relationship between predicted damage intensity (the sum of T-values within building footprints for a given cell) and observed damage intensity (the total area of buildings labeled as damaged in a grid cell) at the level of 500 m grid cells in Gaza. There is strong agreement between the two: especially in Northern Gaza and Khan Younis, the cells that are in the top percentile for predicted damage are also in the top percentile for observed damage.

Treating grid-cell damage purely as a binary classification problem loses information on damage intensity. Yet because the count of damaged buildings per grid-cell contains a large number of zeros and is highly skewed, estimating an ordinary least squares regression model would be inappropriate. A Zero-Inflated Poisson regression model has two parts: a logit model to predict excess zeros, and a Poisson count model:

$$P(Y = y | \mathbf{X}) = \begin{cases} \pi + (1 - \pi) \cdot \exp(-\lambda), & \text{if } y = 0, \\ (1 - \pi) \cdot \frac{\lambda^y \exp(-\lambda)}{y!}, & \text{if } y > 0, \end{cases}$$

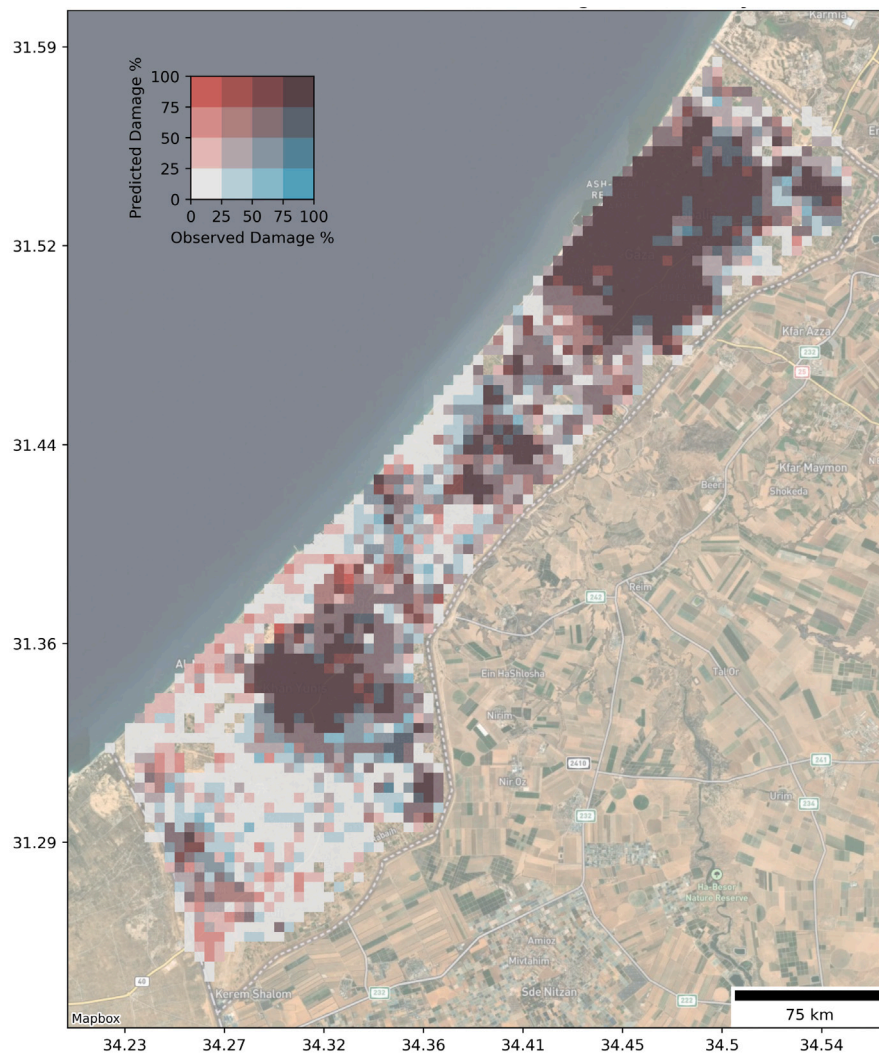
$$\lambda = \exp(\beta_0 + \beta_1 T + \beta_2 B)$$

where  $Y$  is the dependent count variable,  $\mathbf{X}$  is the vector of predictors, which includes the mean value of  $T$  from the PWTT damage probability raster and the total count of buildings  $B$  in a grid cell.  $\pi$  represents the probability of an excess zero from the zero-inflated process. The parameter  $\lambda$  is the Poisson mean, which is modeled as a function of the predictors  $T$  and  $B$ , with their corresponding regression coefficients  $\beta_1$  and  $\beta_2$ . Finally,  $y$  denotes the observed count of damaged buildings in a grid cell, which can take values  $y = 0, 1, 2, \dots$ . This model combines two components: the first is a binary process, where  $y = 0$  with probability  $\pi$ , and the second is a Poisson process, where  $y = 0, 1, 2, \dots$  with mean  $\lambda$  for the non-zero outcomes. The former models the probability of a grid-cell containing no damaged buildings, while the latter models the count of damaged buildings using a Poisson distribution. The results from this specification are reported in Table 5.

In every country, there is a strong positive relationship between the log number of damaged buildings in a grid cell and the mean value of the damage probability raster. Across all cities, a one-unit increase in a grid cell's mean T-value is associated with a 42.8% increase in the number of damaged buildings therein ( $e^{0.356} - 1 = 0.428$ ). The rightmost column under the heading "PWZS" indicates regression results derived from the Pixel-Wise Z-Test, which uses only a single post-conflict image. All coefficients are significant at the 99% confidence level, though there is some range in their magnitudes. These differences reflect the differing spatial patterns of damage across geographic contexts. The average number of damaged buildings per grid cell in Ukraine is 2, while in Palestine it is 100.

R-squared values from the regression models demonstrate a strong association between the damage probability raster and observed damage intensity. Across all cities, over 75% of the variation in the number of damaged buildings in a cell can be explained by the average value of the damage probability raster therein. In rapid-response scenarios, the





**Fig. 8.** Predicted and observed damage intensity, Gaza.

“Observed Damage” refers to the total area of buildings labeled as damaged in a grid cell, while “Predicted Damage” reflects the sum of T-values within building footprints. The percentile distribution is used for both variables. Purple cells indicate cells with both high predicted and observed damage, while white cells indicate low levels of both variables. Blue cells indicate an underprediction of damage intensity, while red cells indicate overprediction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Pixel-Wise Z-Score achieves almost the same performance as the PWTT, with a Pseudo  $R^2$  value just 4% lower.

Framing zonal battle damage assessment as a regression problem has a number of practical benefits for practitioners. Given that over half of the grid-cells in the sample contain at least one damaged building and that some contain over 800, binary damage classification only yields a limited amount of information. Nevertheless, the change detection algorithm elaborated herein achieves an F1 score of 75% for this task. Beyond simply identifying damaged buildings, the PWTT can accurately gauge the intensity of damage within a 2.5 km<sup>2</sup> area across different countries.

#### 4. Discussion

The current state of the art in building damage detection via remote sensing involves the use of deep learning and high resolution optical satellite imagery. Section 3.2 demonstrates that the PWTT achieves an F1 score over 15% higher than the state of the art deep learning model – ChangeOS – on damage detection in Gaza at 6 different time periods. Beyond producing more accurate results, the PWTT solves many of the problems associated with the use of deep learning and high resolution optical satellite imagery, detailed below.

##### 4.1. Data availability

The cost of high-resolution satellite imagery required to run inference using a deep learning model would exceed \$13 million for a single collection of Ukraine (AAAS, 2024). Even if the imagery could be sourced, cloud cover poses a significant challenge given that the annual cloud fraction exceeds 60% in many parts of Ukraine (NASA, 2024). Further atmospheric occlusions in combat zones, such as pyrocumulus clouds and aerosolized dust from rubble, further complicate the use of optical imagery. The SAR imagery used by the PWTT is not only open access but can penetrate through cloud cover, guaranteeing consistent coverage at a high repeat rate.

##### 4.2. Generalization

While neural networks struggle to generalize even within the same country, and often struggle with irregularly shaped buildings as shown in Fig. 6. In contrast, the PWTT conducts unsupervised damage detection by incorporating a year’s worth of historical imagery for each pixel.

Using a validation dataset spanning over 900,000 labeled building footprints in 30 cities and five different countries, the PWTT achieves an overall AUC of 0.87 and F1 score of 65%, using a single threshold value of  $T$  ( $> 3.2$ ). Beyond geographic generalization, the PWTT also maintains robust temporal generalization with AUC values remaining consistent across the 6 time periods analyzed in Gaza. It also achieves an F1 score of 65% for the detection of damage within a 500 m grid cell, and can even generate accurate estimates of damage intensity: the mean value of  $T$  within a grid cell explains 75% of the variation in the number of damaged buildings therein.

Further benchmarking exercises are conducted in [Appendix A. Appendix A.3.2](#) computes the PWTT on the xBD benchmark dataset, and compares the results to four state-of-the-art self-supervised learning methods reported by Xia et al. (2022). These methods include DINO, MoCo v3, ImageNet pre-training, and an asymmetric twin-network architecture with remote-sensing-specific adaptations. The results show that PWTT outperforms all SSL models under low labeling scenarios, but SSL approaches surpass PWTT when 20% of labels are used for fine-tuning. [Appendix A.3.3](#) compares PWTT results to semi-supervised models produced by Google AI, Stanford, and the World Food Programme, on the xBD holdout set. PWTT outperforms the SSL models out of the box, but when a larger number of labels are used to fine-tune the SSL models, they ultimately achieve higher accuracy.

#### 4.3. Resolution

The PWTT maintains Sentinel-1's native spatial resolution of 10m/px, which is considerably lower than the 50 cm that is standard for high resolution imagery. This makes the identification of sparse damage challenging, as well as the precise delineation of damage within dense urban environments.

However, the dominant approach to building damage detection using radar imagery—Coherence Change Detection (CCD)—requires down-sampling to 40m/px, making the PWTT 16 times higher resolution, as shown in [Fig. 1](#). As discussed in [Section 3.3](#), spillover false positives are already a problem at 10 meters per pixel. A comparison between the PWTT and CCD is available in [Table A.10](#). F1 accuracy is on average 9% lower using InSAR, likely due to the effect of spillovers.

#### 5. Conclusion

The generation of public information regarding buildings damaged by conflict has become particularly important in the context of recent, highly destructive wars in Gaza and Ukraine. This paper develops the Pixelwise T-Test (PWTT), a new algorithm for open-access battle damage estimation that is accurate, lightweight, and generalizable. The PWTT addresses many of the problems associated with expense, coverage consistency, and domain shift that affect deep learning based approaches to building damage detection, and outperforms the state of the art deep learning model in Gaza.

These properties enable the creation of interactive Battle Damage Dashboards for [Ukraine](#) and [Gaza](#) that conduct multi-temporal, wide area building damage assessments on the fly using the Google Earth Engine cloud computing platform. These dashboards have three key features: First, they allow for damage assessments to be carried out across an entire country and for any time period during a war. Second, damage estimates are combined with high resolution pre-war population data to estimate the number of people previously living in now-destroyed areas. Third, they integrate additional open source data for further verification in the form of geolocated social media footage of battle damage. Given the contentious nature of battle damage estimates, the use of open-source code and data ensures transparency, explainability, and replicability. Further details on the dashboards are available in [Appendix A](#).

The framework of the Battle Damage Dashboards is highly extensible, supporting modification to fit the needs of stakeholders. With light

modification, alerts could be set up to detect the spread of fighting to different regions in near-real time, and provide early warning to nearby civilians. In a post-war reconstruction scenario, summaries of damage levels by administrative region could be easily generated to target field surveys and ultimate reconstruction efforts.

While maximizing predictive accuracy is a natural goal, it is also incumbent upon researchers to consider the ethics and ultimate goal of battle damage assessment, which is not to fully automate the process. Any model is subject to error, and for high-stakes decision-making around post-conflict reconstruction there must always be a human in the loop. Where more precise damage estimates are required, the PWTT can be deployed to “tip and cue” manual assessments or field surveys. Having a human in the loop is also important in mitigating possible ethical challenges regarding the monitoring of armed conflict; damage assessment is a core workstream for military intelligence agencies tasked with assessing the success of an artillery strike or bombing run, which could be ascertained using the PWTT. As these actors typically have access to higher resolution imagery and lower-latency methods to assess damage to a small target area, it is unlikely that they would rely on the PWTT which is optimized for wide-area monitoring. Nevertheless, researchers must carefully consider the potential benefits and harms of publicly releasing building damage data.

Finally, there are a number of limitations to the approach elaborated herein. Though the accuracy of the PWTT is assessed using over 900,000 building footprints in different climatic regions, further research must test the algorithm in Asia and Latin America. A second area for improvement would be to increase accuracy by testing different change detection algorithms; Bayesian and non-parametric methods in particular hold promise. The creation of an open-access benchmark dataset for battle damage detection comprising over half a million labeled building footprints across 30 cities in four countries provides a common reference for future studies of battle damage.

#### Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ollie Ballinger reports financial support was provided by UK Government Foreign Commonwealth & Development Office (Grant No. GB-GOV-1-300358-413). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

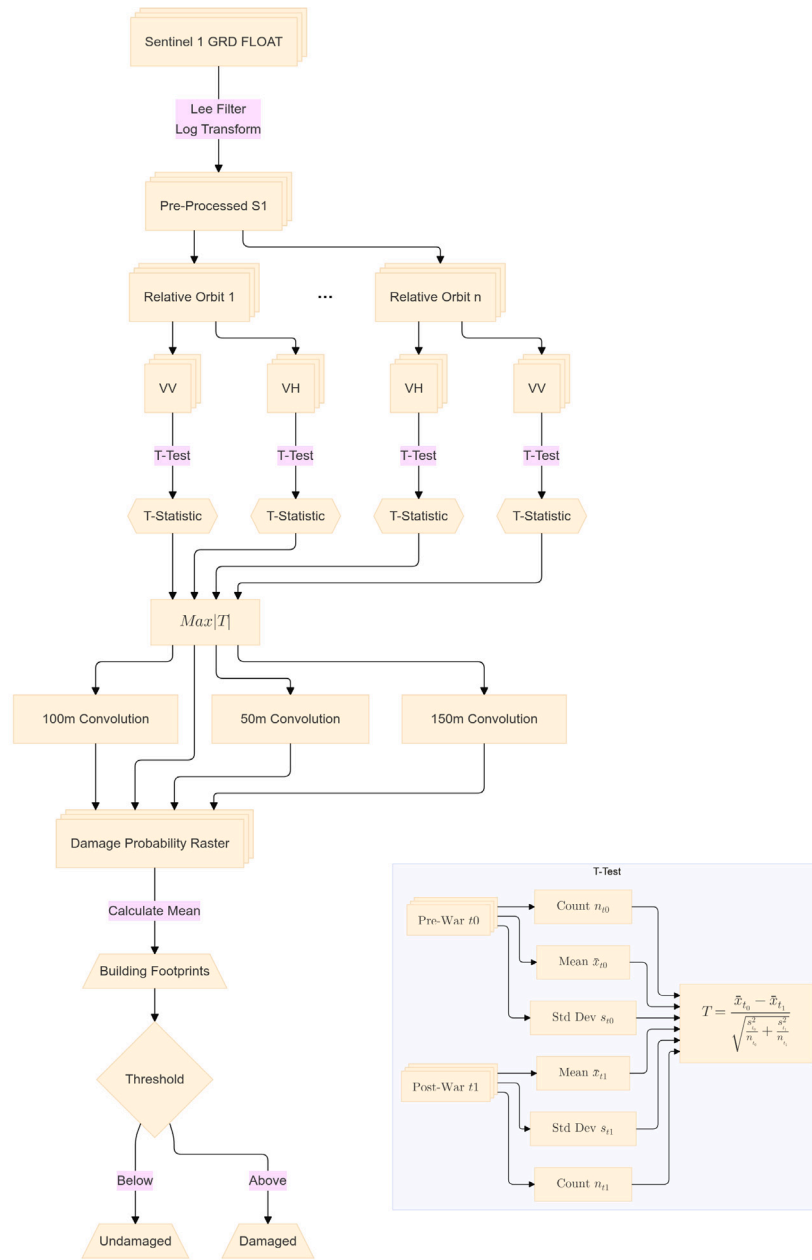
#### Appendix A

##### A.1. Global building footprint datasets

###### A.1.1. Availability

In order to generate building-level damage estimates, the PWTT requires external building footprint data. Such datasets have been proliferating in quantity and quality in recent years. The figure below displays the spatial distribution of two key open source building footprint datasets; the [Microsoft Building Footprints](#) dataset (used in this study) shown in orange, and the “[Open Buildings](#)” dataset published by Google, shown in blue. Combined, these datasets cover 92% of the countries in the world.

A third global building footprint dataset that could be used for building damage detection is [OpenStreetMap](#), though coverage consistency varies as these are derived from community-contributed annotations. A dataset focused on East Asia was also recently released by [Shi et al. \(2023\)](#), and many countries also release their own national building footprint datasets. Finally, there exist datasets that hierarchically combine these different datasets based on reliability in different regions, such as [Overture Maps](#) and [Google-Microsoft Open Buildings - combined by VIDA](#). As such, there are generally very few gaps in coverage, and often multiple different building footprint datasets are available for a given location.



**Fig. A.1.** Pixel wise T-Test process diagram.

Image rasters are displayed as rectangles, image collections are displayed as stacked rectangles, building footprints are shown as trapezoids. A separate set of two T-Tests is conducted for each unique Relative Orbit, of which there can be multiple.

#### A.1.2. Reliability

Though building footprint datasets are widely available, they must also be reliable in order to produce accurate building damage estimates. Table A.1 reports the accuracy statistics of the Microsoft Buildings dataset, used herein. In Europe and the Middle East, where all but one of the AOIs used for evaluation in this study are located, the F1 score for building localization is 90%:

This is 5% higher than the reported F1 score for building localization in the ChangeOS building damage detection framework by Zheng et al. (2021), which is 85%. Furthermore, as shown in Fig. 6, the latter struggles to detect buildings in new geographies such as Gaza. As such, the building footprints used by the PWTT are of similar if not higher quality in some regions than those produced by damage detection models which generate their own footprints.

One of the criteria for including a particular city in the evaluation dataset of this study is the availability of reliable building footprint

data. A detailed manual review of each city was conducted and the building footprints were found to be of high quality, consistent with the reported accuracy statistics in the table above. Fig. A.3 plots Microsoft Building footprints in Yei, Sudan. According to the metrics in Table A.1, the building footprints in Africa are of the lowest quality, with an F1 score nearly 10% lower than Europe and the Middle East. Still, the footprints align closely with the buildings visible in the imagery, across a range of different sizes, densities, and roofing materials. Thus, under normal circumstances, even the lowest quality building footprints are reliable enough for damage assessment purposes.

However, under certain specific circumstances, some locations may have degraded building footprints due to destruction that occurred before or during the period in which the source imagery for footprint extraction was collected. This is the case in some Syrian cities discussed in Appendix A.3.4; Fig. A.4 shows the Masaa al-Arbreen neighborhood



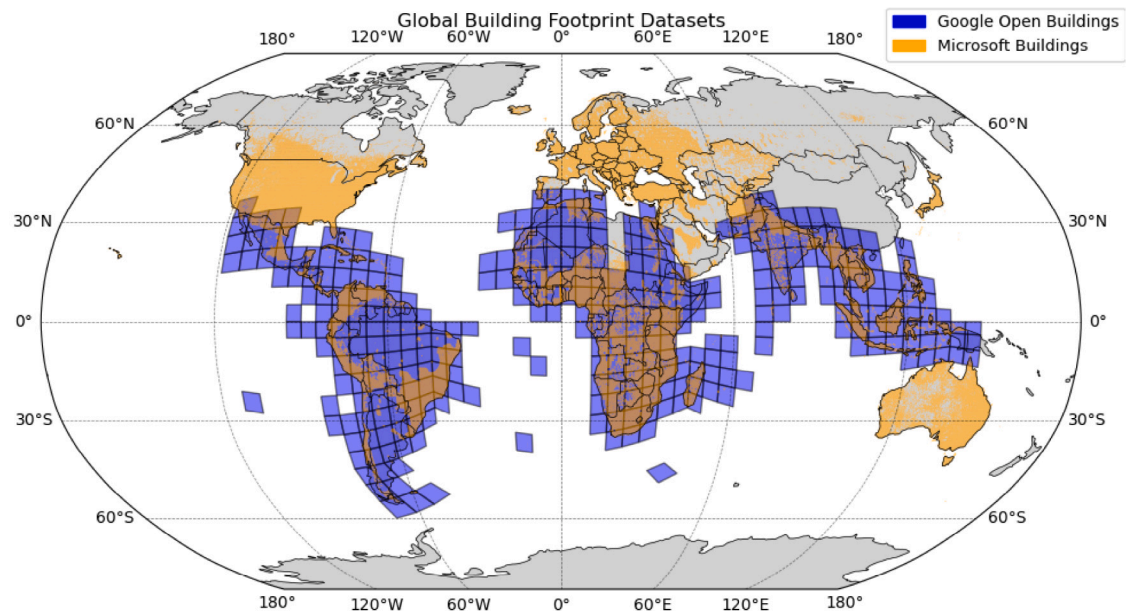


Fig. A.2. Global distribution of open-access building footprint datasets.

**Table A.1**  
Microsoft buildings regional accuracy statistics.

| Region        | Precision | Recall | F1    | IoU   | Rotation error [deg] |
|---------------|-----------|--------|-------|-------|----------------------|
| Africa        | 94.4%     | 70.9%  | 80.8% | 64.5% | 5.67                 |
| Caribbean     | 92.2%     | 76.8%  | 83.8% | 64.0% | 6.64                 |
| Central Asia  | 97.17%    | 79.47% | 87.4% | 68.2% | 6.91                 |
| Europe        | 94.3%     | 85.9%  | 89.9% | 65.1% | 10.28                |
| Middle East   | 95.7%     | 85.4%  | 90.2% | 65.1% | 9.30                 |
| South America | 95.4%     | 78.0%  | 85.8% | 66.7% | 6.34                 |
| South Asia    | 94.8%     | 76.7%  | 84.9% | 63.1% | 6.25                 |

Accuracy statistics were sourced from the Microsoft Buildings [GitHub page](#). F1 score is based on the author's calculation.

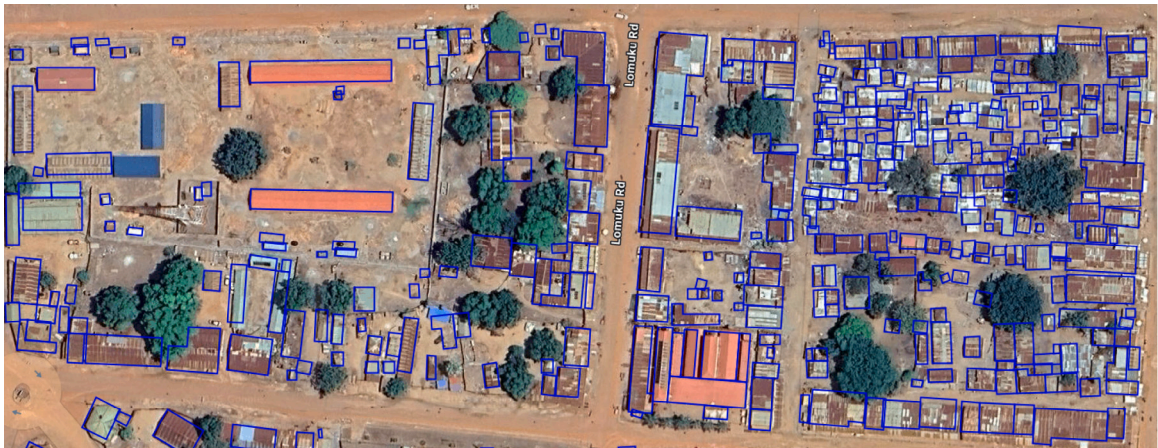


Fig. A.3. Building footprints in Yei, Sudan. Microsoft Building footprints overlaid on high resolution optical satellite imagery of Yei, Sudan.

in Hama, Syria, with building footprint outlines shown in blue, and UNOSAT damage annotations shown in red. The damage to this neighborhood – which was razed by bombings in 2012 – seemingly predates the imagery used to generate the building footprints. As such, cities such as Hama and Daraa were excluded from the main analysis on the basis of having unreliable building footprints, as determined by manual review.

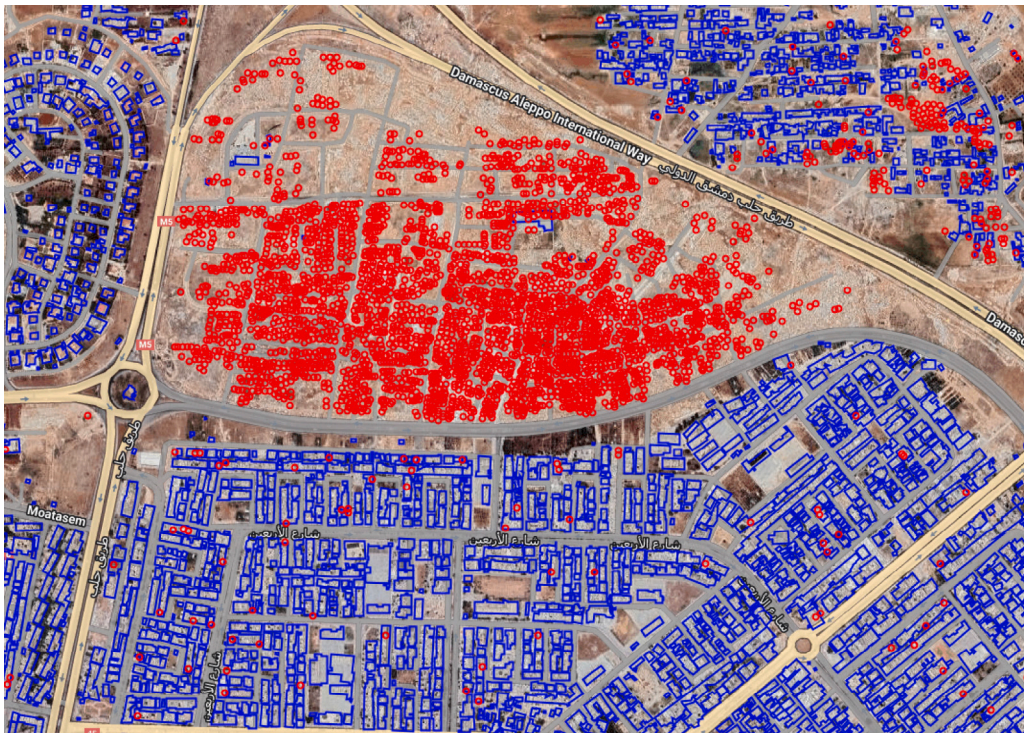
Circumstances such as these are exceptional, and were not found to be common in other areas. Nevertheless, it is important that a manual review of outputs is conducted, which includes both quantitatively and

qualitatively assessing the building footprints used in the analysis. Yet even in locations where building footprints are missing or degraded, a zonal damage assessment can still be conducted via the PWTT, as shown in Fig. 8.

A.2. Battle damage dashboards

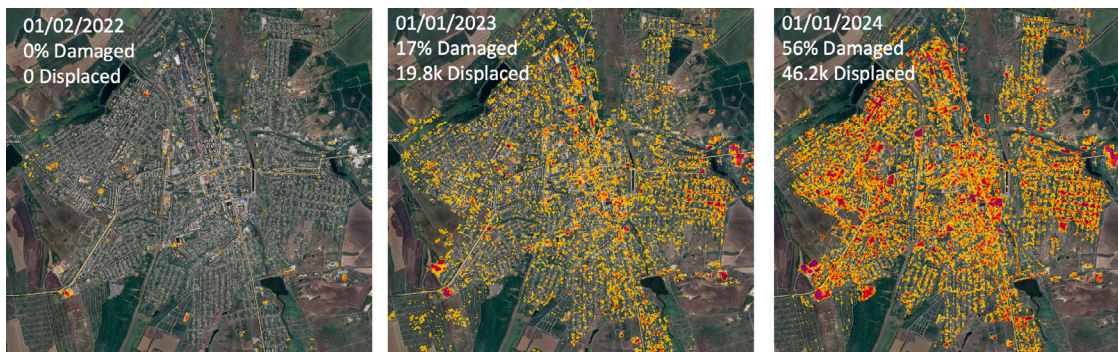
See Fig. A.5.





**Fig. A.4.** Building footprints in Hama, Syria.

Microsoft Building footprints overlaid on high resolution optical satellite imagery of Hama, Syria, with UNOSAT damage points shown in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. A.5.** Ukraine battle damage dashboard analysis of Bakhmut.

Building damage and population displacement estimates for Bakhmut generated at three time intervals for the city of Bakhmut, Ukraine, using the Battle Damage Dashboard.

#### A.2.1. Damage estimates

The core feature of the Battle Damage Dashboard is that it enables building damage assessment to be carried out across an entire country over the full duration of a conflict. A date slider allows users to conduct damage detection for any two-month period during the war, and damaged building footprints can be downloaded within a user-defined area.

Building-level precision and recall are critical measures of model performance for humanitarian practitioners, as they provide highly interpretable measures of uncertainty. Country-wide precision-recall curves at the building level are used in the visualization and analysis of estimated damage. The value of  $T$  that maximizes F1 accuracy is used as the cutoff value for damage prediction in each country. Areas with lower  $T$  values are masked, and areas with higher values are colored using a color scale. Corresponding precision values are communicated for the color scale. For example in the Ukraine Battle Damage Dashboard, a yellow pixel has a 60% chance of being damaged, while a purple area has an  $> 83\%$  chance of being damaged.

#### A.2.2. Population exposure

Damage predictions are combined with population density estimates from Oak Ridge National Laboratory's LandScan HD program to quantify the number of people likely to be permanently displaced by battle damage. LandScan data has been used extensively in the literature on disaster response to estimate populations affected by natural hazards and conflict (Laverdiere et al., 2022; Urban et al., 2023). Ambient population distributions are estimated by combining high resolution building features, occupancy data, land use, and data on Points of Interest (Urban et al., 2023). LandScanHD data are made available at a spatial resolution of 90 m. Once damaged areas have been identified via the PWTT, the ambient pre-war population in the damaged area is calculated using LandScan data. Importantly, these data reflect the pre-war population distribution and as such they reflect the number of people previously living in now-destroyed areas.

This provides an important additional layer of information for humanitarian practitioners, enabling more precise needs assessments

**Table A.2**

Building-level damage detection accuracy using the continuous change detection and classification (CCDC) algorithm.

| Country   | Location        | AUC   | Accuracy | F1    | Precision | Recall | N       |
|-----------|-----------------|-------|----------|-------|-----------|--------|---------|
| Palestine | Gaza            | 64.41 | 67.25    | 78.44 | 66.51     | 95.6   | 228 728 |
| Ukraine   | Kherson         | 64.3  | 93.21    | 10.14 | 8.97      | 11.66  | 7302    |
| Ukraine   | Antonivka       | 77.5  | 95.76    | 5.73  | 16.7      | 3.46   | 905     |
| Ukraine   | Lysychansk      | 78.22 | 80.89    | 48.11 | 43.45     | 53.9   | 20 246  |
| Ukraine   | Avdiivka        | 67.75 | 75.74    | 41.02 | 38.1      | 44.42  | 7262    |
| Ukraine   | Melitopol       | 72.74 | 87.31    | 1.62  | 0.87      | 12.58  | 32 373  |
| Ukraine   | Sievierodonetsk | 70.29 | 60.96    | 62.47 | 52.1      | 77.99  | 5970    |
| Ukraine   | Kramatorsk      | 86.25 | 96.88    | 42.44 | 34.31     | 55.62  | 21 880  |
| Ukraine   | Mykolaiv        | 83.44 | 98.28    | 27.43 | 23.42     | 33.1   | 60 467  |
| Ukraine   | Rubizhne        | 78.6  | 68.58    | 67.29 | 57.32     | 81.46  | 8899    |
| Ukraine   | Sumy            | 73.37 | 94.83    | 0.98  | 0.51      | 9.7    | 28 265  |
| Ukraine   | Shchastia       | 84.91 | 95.89    | 43.29 | 38.59     | 49.31  | 1293    |
| Ukraine   | Kremenchuk      | 86.13 | 99.03    | 1.22  | 1.57      | 1.0    | 27 497  |
| Ukraine   | Kharkiv         | 81.55 | 97.01    | 26.48 | 26.27     | 26.68  | 107 976 |
| Ukraine   | Mariupol        | 72.52 | 66.09    | 70.35 | 61.15     | 82.82  | 18 446  |
| Ukraine   | Borodyanka      | 86.44 | 92.11    | 49.41 | 38.67     | 68.41  | 3912    |
| Ukraine   | Volnovakha      | 72.87 | 80.7     | 44.07 | 42.39     | 45.89  | 8616    |
| Ukraine   | Irpin           | 80.77 | 83.71    | 57.24 | 54.97     | 59.71  | 7242    |
| Ukraine   | Bucha           | 85.79 | 98.1     | 70.33 | 97.11     | 55.13  | 5739    |
| Ukraine   | Hostomel        | 81.02 | 77.19    | 62.53 | 59.03     | 66.46  | 4175    |
| Ukraine   | Vorzel          | 78.3  | 86.32    | 7.89  | 5.13      | 17.05  | 959     |
| Ukraine   | Moschun         | 61.9  | 57.77    | 72.79 | 57.28     | 99.83  | 603     |
| Ukraine   | Okhtyrka        | 73.44 | 94.96    | 16.02 | 10.74     | 31.52  | 15 905  |
| Ukraine   | Trostanets      | 50.0  | 97.2     | 0.0   | 0.0       | 0.0    | 8913    |
| Ukraine   | Chernihiv       | 87.0  | 94.7     | 51.26 | 55.65     | 47.51  | 29 929  |
| Ukraine   | Makariv         | 73.98 | 92.88    | 32.24 | 45.22     | 25.05  | 3514    |
| Syria     | Aleppo          | 70.31 | 57.87    | 57.28 | 43.76     | 82.87  | 65 870  |
| Syria     | Raqqa           | 72.74 | 64.48    | 72.52 | 62.02     | 87.3   | 24 689  |
| Iraq      | Mosul           | 74.98 | 79.17    | 39.24 | 32.04     | 50.61  | 137 794 |
| Sudan     | Yei             | 50.0  | 98.39    | 0.0   | 0.0       | 0.0    | 9608    |
| All       | All             | 84.05 | 77.02    | 61.36 | 51.6      | 75.69  | 904 977 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. Damage estimates were computed using the Continuous Change Detection and Classification (CCDC) Algorithm.

based on whether fighting is predominantly occurring in industrial zones, high-density urban areas, or suburban neighborhoods. For example, the town of Popasna in Ukraine sustained heavy damage, resulting in over 1000 damaged buildings – largely single family homes – with a corresponding pre-war population of around 8000. An equivalent number of damaged buildings in Gaza City’s Rimal neighborhood permanently displaces roughly 40000 people. The addition of population density data allows the Battle Damage Dashboards to overcome a key limitation of using the number, proportion, or total area of damaged buildings in humanitarian response.

#### A.2.3. Geolocated social media footage

While all accuracy statistics are calculated on the basis of UNOSAT damage labels, these are only available for a handful of cities. As such, an additional source of verification data is provided in the form of geolocated footage from social media showing battle damage. These are sourced from the Centre for Information Resilience, an organization which maintains a crowdsourced database of photos and videos of conflict events extracted from social media platforms such as Telegram, TikTok, and Twitter (Strick, 2023). The source media is archived, chrono- and geolocated, categorized, and finally reviewed by a team of investigators. In the battle damage dashboards, clicking on a geolocated event will open a panel showing a brief description of the event, the date, a link to the source media, and a link to the geolocation of the event. Despite the review process, it should be noted that geolocated photos and videos from social media are subject to some level of error and should not be treated as ground truth. Nevertheless, they provide an important additional source of corroborative evidence in areas where UNOSAT damage labels are missing.

### A.3. Benchmarking

#### A.3.1. Unsupervised change detection

Because the PWTT is an unsupervised change detection algorithm, it is appropriate to compare its results to existing methods. One of the most highly cited and widely used algorithms of this kind is the Continuous Change Detection and Classification (CCDC) algorithm developed by [Zhu and Woodcock \(2014\)](#) for land cover change detection in Landsat imagery. CCDC is a timeseries model which flexibly models intra-annual change (seasonality), gradual inter-annual change (e.g. vegetative growth), and distinguishes these from abrupt changes using the following functional form:

$$\hat{\rho}(i, x)_{OLS} = a_{0,i} + a_{1,i} \cos\left(\frac{2\pi}{T}x\right) + b_{1,i} \sin\left(\frac{2\pi}{T}x\right) + c_{1,i}x \quad \{\tau_{k-1}^* < x \leq \tau_k^*\}$$

Where  $x$  is the date,  $i$  is the  $i$ th Landsat Band, and  $T$  is the number of days per year ( $T = 365$ ). The coefficient  $a_{0,i}$  represents the overall value for the  $i$ th Landsat Band;  $a_{1,i}$  and  $b_{1,i}$  represent intra-annual change;  $c_{1,i}$  is the coefficient for inter-annual change;  $\tau_k^*$  denotes the  $k$ th break point; and  $\hat{\rho}(i, x)_{OLS}$  is the predicted value for the  $i$ th Landsat Band at date  $x$ .

Though CCDC was developed for optical imagery and performs particularly well at identifying changes in vegetative land cover, building damage detection is effectively a break point identification task, as shown in [Fig. 2](#). A further advantage of applying CCDC to building damage detection is that it does not require manually specifying a reference or inference period, as it identifies breakpoints over the entire period as well as their timing and magnitude.

[Table A.2](#) applies CCDC to the benchmark building damage detection dataset developed in this paper. The same time periods as the PWTT are employed: 12 months preceding and one month following the UNOSAT damage annotations. The CCDC is applied to both the VV



**Table A.3**

F1 score comparison with self-supervised learning methods on the xBD dataset.

| Labeling | Method     | Damage | No damage | Minor | Major | Destroyed |
|----------|------------|--------|-----------|-------|-------|-----------|
| 1%       | ImageNet   | 0.321  | 0.387     | 0.136 | 0.234 | NaN       |
|          | DINO       | 0.366  | 0.480     | 0.157 | 0.439 | NaN       |
|          | MoCo v3    | 0.379  | 0.425     | 0.124 | 0.337 | NaN       |
|          | ATN        | 0.390  | 0.486     | 0.261 | 0.345 | NaN       |
| 20%      | ImageNet   | 0.587  | 0.604     | 0.278 | 0.471 | 0.456     |
|          | DINO       | 0.601  | 0.667     | 0.229 | 0.384 | 0.447     |
|          | MoCo v3    | 0.639  | 0.562     | 0.230 | 0.392 | 0.400     |
|          | ATN        | 0.636  | 0.646     | 0.314 | 0.480 | 0.380     |
| –        | PWTT       | 0.476  | 0.842     | 0.276 | 0.144 | 0.304     |
| –        | PWTT + XGB | 0.672  | 0.894     | 0.512 | 0.338 | 0.600     |

This table is a reproduction of Table 1 in [Xia et al. \(2022\)](#). ATN refers to the Asymmetric Twin Network model elaborated in the aforementioned paper. Accuracy assessment is carried out on the holdout set from the xBD dataset. The values reported in the table are the F1 scores for the damage category indicated by the column. PWTT damage classification is conducted by selecting a threshold value of T using a precision–recall curve. The PWTT + XGB row reports the results of a gradient boosting model trained on the PWTT output using 5-fold cross-validation.

**Table A.4**

Performance on the Santa Rosa dataset with different methods and labeled data sizes.

| Approach | # Labeled data | Twin tower  | Fully supervised | MixMatch    | FixMatch    |
|----------|----------------|-------------|------------------|-------------|-------------|
| SSL      | 500            | 0.69 ± 0.03 | 0.96 ± 0.01      | 0.96 ± 0.01 | 0.98 ± 0.00 |
|          | 100            | 0.58 ± 0.04 | 0.92 ± 0.01      | 0.88 ± 0.04 | 0.97 ± 0.01 |
|          | 50             | 0.54 ± 0.03 | 0.91 ± 0.03      | 0.85 ± 0.08 | 0.96 ± 0.02 |
|          | 10             | 0.54 ± 0.03 | 0.74 ± 0.08      | 0.70 ± 0.06 | 0.92 ± 0.07 |
| PWTT     | 0              | <b>0.90</b> | <b>0.90</b>      | <b>0.90</b> | <b>0.90</b> |

**Table A.5**Sample comparison with [Mueller et al. \(2021\)](#).

| City        | <a href="#">Mueller et al. (2021)</a> |             | PWTT    |             |
|-------------|---------------------------------------|-------------|---------|-------------|
|             | Patches                               | % Destroyed | Patches | % Destroyed |
| Aleppo      | 95,746                                | 1.83        | 108,337 | 1.58        |
| Daraa       | 15,573                                | 1.00        | 10,999  | 1.00        |
| Deir-Ez-Zor | 14,086                                | 2.86        | 15,185  | 0.88        |
| Hama        | 31,673                                | 3.73        | 21,670  | 0.35        |
| Raqqa       | 22,523                                | 1.96        | 32,121  | 7.86        |
| Homs        | 40,007                                | 8.26        | –       | –           |
| Damascus    | –                                     | –           | 105,459 | 3.81        |
| Idlib       | –                                     | –           | 5,426   | 1.31        |

This table compares the samples used in the study by Mueller et. al. 2021, with the reproduction used in this study. The values under the Mueller et. al. heading were calculated from Table 1 of the study; “% Destroyed” corresponds to column 5 of that table, while the number of patches was calculated by dividing column 2 by column 1.

**Table A.6**Comparison of damage assessment to Syrian cities with [Mueller et al. \(2021\)](#).

| Location    | PWTT         |              |           | <a href="#">Mueller et al. (2021)</a> |             |                         |           |
|-------------|--------------|--------------|-----------|---------------------------------------|-------------|-------------------------|-----------|
|             | AUC          | Unbalanced   | Balanced  | Trained on Aleppo only                |             | Train/test on same city |           |
|             |              | Precision    | Precision | AUC                                   | Precision   | AUC                     | Precision |
| Raqqa       | <b>74.29</b> | <b>18.67</b> | 70.56     | 58.1                                  | 2.7         | 87.6                    | 32.1      |
| Daraa       | 71.27        | 2.94         | 70.48     | <b>84.5</b>                           | <b>13.6</b> | 89.0                    | 11.7      |
| Aleppo      | 74.03        | 4.51         | 71.90     | <b>89.6</b>                           | <b>33.3</b> | 91.5                    | 35.7      |
| Idlib       | 67.19        | 2.93         | 67.79     | –                                     | –           | –                       | –         |
| Hama        | 61.08        | 0.60         | 58.68     | <b>79.5</b>                           | <b>32.5</b> | 91.0                    | 68.0      |
| Damascus    | 65.99        | 5.82         | 60.24     | –                                     | –           | –                       | –         |
| Deir-Ez-Zor | <b>77.49</b> | <b>12.09</b> | 81.68     | 65.5                                  | 6.5         | 80.0                    | 21.7      |
| Homs        | –            | –            | –         | 63.6                                  | 18.3        | 85.7                    | 55.2      |
| All         | 65.63        | 7.88         | 67.73     | –                                     | –           | 90.7                    | 42.5      |

The columns under the PWTT heading calculate AUC for the full unbalanced sample, and average precision on both the unbalanced and balanced samples. The columns labeled “Train/test on same city” refer to Table 2 in [Mueller et al. \(2021\)](#), while the columns “Trained on Aleppo only” are taken from Table S3 in the appendix. In both cases, these refer to the results from the full model (the first stage convolutional neural network with the second-stage random forest with spatial and temporal leads and lags), and Precision refers to average precision on the unbalanced sample.

and VH bands of the Sentinel-1 imagery, yielding a pixel-wise change probability value comparable to the PWTT’s T-statistic.

The CCDC demonstrates strong performance on the task of building damage detection; AUC and F1 for the full sample are around 3% lower than the PWTT. In certain locations, CCDC outperforms the PWTT

by a few points. However, the CCDC’s performance is generally more variable; in Kherson, AUC is around 30% lower than the PWTT, and in Gaza it is 12% lower. In Trostianets and Yei, Sudan, it fails altogether.

Nevertheless, these are strong results for a framework designed for a very different input modality and domain. The unsupervised nature

**Table A.7**

Building-level damage detection accuracy using a single post-war image, via pixel-wise Z-score (PWZS)

| country   | Location        | AUC   | Accuracy | F1    | Precision | Recall | N       |
|-----------|-----------------|-------|----------|-------|-----------|--------|---------|
| Palestine | Gaza            | 75.81 | 72.57    | 80.77 | 71.75     | 92.38  | 228 728 |
| Ukraine   | Lysychansk      | 70.08 | 76.74    | 40.54 | 34.97     | 48.22  | 20 246  |
| Ukraine   | Avdiivka        | 62.86 | 75.12    | 36.2  | 35.28     | 37.17  | 7262    |
| Ukraine   | Melitopol       | 70.89 | 82.49    | 5.3   | 2.78      | 58.91  | 32 373  |
| Ukraine   | Sievierodonetsk | 65.07 | 53.46    | 60.7  | 46.82     | 86.26  | 5970    |
| Ukraine   | Kramatorsk      | 77.86 | 95.84    | 29.14 | 22.49     | 41.37  | 21 880  |
| Ukraine   | Mykolaiv        | 73.38 | 97.89    | 21.95 | 17.22     | 30.25  | 60 467  |
| Ukraine   | Rubizhne        | 73.6  | 66.18    | 63.25 | 55.59     | 73.35  | 8899    |
| Ukraine   | Sumy            | 53.29 | 99.32    | 0.0   | 0.0       | 0.0    | 28 265  |
| Ukraine   | Shchastia       | 74.15 | 96.65    | 24.67 | 43.45     | 17.22  | 1293    |
| Ukraine   | Kremenchuk      | 83.06 | 99.08    | 28.06 | 26.33     | 30.04  | 27 497  |
| Ukraine   | Kharkiv         | 74.47 | 97.26    | 23.95 | 27.23     | 21.37  | 107 976 |
| Ukraine   | Mariupol        | 70.12 | 61.63    | 68.86 | 56.84     | 87.35  | 18 446  |
| Ukraine   | Borodyanka      | 82.98 | 92.04    | 41.26 | 35.33     | 49.59  | 3912    |
| Ukraine   | Volnovakha      | 70.05 | 79.34    | 42.12 | 39.31     | 45.37  | 8616    |
| Ukraine   | Bucha           | 85.97 | 98.13    | 70.57 | 98.56     | 54.96  | 5739    |
| Ukraine   | Hostomel        | 82.04 | 76.1     | 63.73 | 56.35     | 73.35  | 4175    |
| Ukraine   | Vorzel          | 66.89 | 97.05    | 24.88 | 100.0     | 14.21  | 959     |
| Ukraine   | Moschun         | 61.42 | 58.29    | 72.59 | 57.78     | 97.6   | 603     |
| Ukraine   | Irpin           | 77.13 | 81.48    | 49.16 | 49.29     | 49.02  | 7242    |
| Ukraine   | Trostanets      | 88.02 | 98.43    | 62.99 | 92.64     | 47.72  | 8913    |
| Ukraine   | Okhtyrka        | 85.88 | 98.56    | 49.99 | 53.27     | 47.08  | 15 905  |
| Ukraine   | Chernihiv       | 76.32 | 92.11    | 34.7  | 33.74     | 35.71  | 29 929  |
| Ukraine   | Makariv         | 72.37 | 89.37    | 27.04 | 25.23     | 29.12  | 3514    |
| Iraq      | Mosul           | 70.53 | 79.06    | 35.64 | 30.12     | 43.65  | 137 794 |
| Sudan     | Yei             | 72.78 | 96.81    | 13.49 | 11.95     | 15.49  | 9608    |
| Syria     | Raqqa           | 66.23 | 56.7     | 70.55 | 55.56     | 96.59  | 24 689  |
| Syria     | Aleppo          | 62.31 | 50.29    | 52.6  | 38.96     | 80.91  | 65 870  |
| All       | All             | 81.73 | 75.98    | 58.43 | 50.88     | 68.62  | 896 770 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. A Pixel-Wise Z-Score (PWZS) is calculated using the first available post-conflict image, by subtracting the pre-war mean from the post-event image and dividing by the pre-war standard deviation. A cutoff for binary damage classification is selected empirically using a precision–recall curve.



**Fig. A.6.** Destruction of the Jobar neighborhood of Damascus, Syria in August 2014.

Screenshot of Google Earth from August 2014, at approximately 33.541159, 36.346802.

of the CCDC, as well as its ability to detect the timing of changes makes it highly suited to this domain. Break point detection in Sentinel-1 timeseries for building damage detection is thus a promising avenue for further research.

### A.3.2. Self-supervised learning

A recent development in the field of building damage detection via remote sensing involves the application of Self-Supervised Learning (SSL), rather than traditional supervised learning. SSL involves training a model using pseudo-labeling, often by using a weakly augmented version of the unlabeled data and calculating loss based on the model's ability to reconstruct the ground truth image. SSL has a number of advantages in this domain, as it can yield performance matching that of supervised models using a far smaller quantity of labeled data by fine tuning large generalist pre-trained networks for specific downstream tasks.

Xia et al. (2022) conduct a comprehensive study assessing the performance of multiple different state of the art SSL pre-training strategies on building damage detection on the xBD benchmark dataset. These include Self-Distillation with No Labels (DINO) (Caron et al., 2021), in which a student model is trained to match a teacher model's output using augmented input imagery, and Momentum Contrast v3 (MoCo v3) (Chen et al., 2021), which extends contrastive learning to vision transformers. They also develop a novel approach based on an asymmetric twin network architecture and a modified DINO pre-training framework with remote sensing domain-specific adaptations. Finally, they compare all three of these methods to a model initialized with ImageNet weights. Table A.3 compares the results of the PWTT to the results of different SSL approaches reported in Table 1 of Xia et al. (2022).

In all cases, the SSL models require some labeled data for fine-tuning on the task of building damage detection. Under low-labeling conditions (1% of the labeled data), the PWTT outperforms all SSL methods

**Table A.8**  
Building-level damage detection accuracy using a balanced sample.

| country   | Location        | AUC   | Accuracy | F1    | Precision | Recall | N       |
|-----------|-----------------|-------|----------|-------|-----------|--------|---------|
| Palestine | Gaza            | 78.18 | 68.9     | 68.29 | 57.0      | 85.14  | 328 974 |
| Ukraine   | Kherson         | 94.18 | 45.26    | 56.5  | 97.78     | 39.73  | 60      |
| Ukraine   | Antonivka       | 49.73 | 41.46    | 57.57 | 76.98     | 45.97  | 18      |
| Ukraine   | Lysychansk      | 75.65 | 71.45    | 81.99 | 73.13     | 93.28  | 3006    |
| Ukraine   | Avdiivka        | 66.65 | 74.75    | 85.43 | 74.63     | 99.9   | 1264    |
| Ukraine   | Melitopol       | 81.01 | 80.13    | 88.53 | 79.95     | 99.16  | 152     |
| Ukraine   | Sievierodonetsk | 73.18 | 72.14    | 82.63 | 72.14     | 96.68  | 2904    |
| Ukraine   | Kramatorsk      | 93.08 | 94.52    | 96.85 | 96.03     | 97.69  | 98      |
| Ukraine   | Mykolaiv        | 84.99 | 81.15    | 87.52 | 86.77     | 88.28  | 406     |
| Ukraine   | Rubizhne        | 78.8  | 71.8     | 77.74 | 71.35     | 85.39  | 5992    |
| Ukraine   | Sumy            | 72.72 | 72.63    | 76.76 | 85.61     | 69.58  | 46      |
| Ukraine   | Shchastia       | 70.5  | 59.59    | 73.71 | 61.02     | 93.06  | 50      |
| Ukraine   | Kremenchuk      | 98.95 | 98.01    | 98.83 | 98.59     | 99.06  | 60      |
| Ukraine   | Kharkiv         | 82.16 | 78.74    | 86.77 | 79.61     | 95.34  | 1842    |
| Ukraine   | Mariupol        | 70.51 | 70.82    | 81.9  | 70.4      | 97.88  | 11 592  |
| Ukraine   | Borodyanka      | 82.29 | 79.66    | 80.71 | 84.78     | 77.02  | 306     |
| Ukraine   | Volnovakha      | 75.25 | 68.6     | 75.47 | 68.37     | 84.22  | 2094    |
| Ukraine   | Bucha           | 91.54 | 83.02    | 88.52 | 82.95     | 94.9   | 184     |
| Ukraine   | Hostomel        | 86.9  | 82.7     | 88.33 | 86.31     | 90.45  | 1172    |
| Ukraine   | Vorzel          | 65.17 | 45.29    | 57.6  | 64.03     | 52.34  | 22      |
| Ukraine   | Moschun         | 71.71 | 62.55    | 70.29 | 58.6      | 87.81  | 668     |
| Ukraine   | Irpri           | 85.0  | 78.89    | 83.82 | 81.27     | 86.53  | 1634    |
| Ukraine   | Trostanets      | 85.94 | 79.22    | 88.08 | 81.15     | 96.31  | 128     |
| Ukraine   | Okhtyrka        | 89.23 | 85.47    | 90.46 | 91.13     | 89.79  | 128     |
| Ukraine   | Chernihiv       | 82.26 | 75.95    | 83.71 | 77.17     | 91.46  | 1944    |
| Ukraine   | Makariv         | 87.67 | 77.66    | 84.1  | 75.62     | 94.71  | 282     |
| Syria     | Raqqa           | 76.34 | 70.35    | 77.72 | 70.07     | 87.26  | 22 398  |
| Syria     | Aleppo          | 74.58 | 71.1     | 77.49 | 70.58     | 85.88  | 40 596  |
| Sudan     | Yei             | 81.54 | 70.33    | 71.18 | 62.16     | 83.27  | 378     |
| Iraq      | Mosul           | 75.99 | 67.7     | 73.24 | 66.54     | 81.44  | 31 720  |
| All       | All             | 87.05 | 77.62    | 79.65 | 71.72     | 89.55  | 460 118 |

Building-level accuracy statistics are generated using UNOSAT damage annotations joined with building footprints. For each location, a threshold value of T is selected using a precision–recall curve to generate a binary damage classification. All statistics are weighted by building area. Balanced samples were generated by randomly sampling the same number of undamaged buildings as damaged buildings for each location.

for all categories of damage except the identification of major damage. The SSL models fail to produce estimates for the “Destroyed” category altogether. When 20% of the labeled data is used for fine-tuning, PWTT manages to outperform all SSL models in the “No damage” category and remains competitive in the identification of minor damage, outperforming DINO and MoCo v3, but is overtaken by the SSL methods in the remaining categories.

It is worth noting that given the size of the xBD dataset, using 20% of the labels may seem like a relatively low labeling scenario, but this still amounts to around 170,000 labels. For illustrative purposes, the PWTT+XGB row reports accuracy statistics when a lightweight gradient boosting model is trained on top of the PWTT output using 5-fold cross-validation. This model outperforms all of the SSL methods across all damage categories except “Major damage”, under a 20% labeling scenario. However, the most realistic building damage detection scenario – whether in the wake of natural disaster or war – is the complete absence of labels, hence the need for automated methods.

Furthermore, as discussed in Section 1, there are important paradigmatic differences between natural disaster and conflict-induced building damage assessments; the former involves a single destructive event in a relatively small area, while the latter requires wide-area, multitemporal monitoring, which the PWTT is optimized for. Nevertheless, the PWTT outperforms cutting edge self-supervised learning methods under low-labeling scenarios on the standard building damage detection benchmark dataset.

### A.3.3. Semi-supervised learning

As the name suggests, Semi-Supervised Learning combines the use of pseudo-labels (of the kind used in self-supervised learning), as well as actual labels. A recent paper produced by a collaboration between Google AI, Stanford, and the UN World Food Programme evaluated a

number of Semi-Supervised Learning techniques on the task of building damage detection (Lee et al., 2020). They evaluate four models: a Twin Tower Convolutional Neural Network model, which performed best in the past work by Xu et al. (2019), a fully-supervised model with only labeled training data, and two semi-supervised architectures, MixMatch and FixMatch. They test these architectures on multiple disasters. Direct benchmarking is only possible for one of these – the Santa Rosa wildfire – as it is part of the publicly available XBD dataset.

Table A.4 corresponds to Table 1 in Lee et al. (2020), and reports the results of evaluation on the Santa Rosa wildfire dataset with incrementally larger amounts of training data. An additional row is added for the PWTT, which achieves accuracy of 90% out of the box.

The PWTT outperforms three out of four models and ties with FixMatch in low-labeling scenarios (10 labels), and outperforms Twin Tower – the best-performing model in Xu et al. (2019), for all increments of training data. However, when larger quantities of labeled data are used to fine-tune the models, they outperform the PWTT.

Semi-Supervised Learning models represent a promising avenue for further research in the field of building damage detection; However, the models used by Lee et al. (2020) still require manual fine-tuning, as well as high-resolution optical satellite imagery which is cost-prohibitive and susceptible to atmospheric occlusions. While these factors may not be a problem for natural disasters, which often involve a single shock affecting a relatively small area, they severely limit these models’ utility for wide-area, multi-temporal conflict damage monitoring.

### A.3.4. The Syrian civil war

This section conducts a detailed comparison between the results of the Pixel-Wise T-Test and the results from the study by Mueller et al. (2021) on the Syrian civil war. To replicate the design of the



**Table A.9**  
500 m grid-cell level damage detection accuracy.

| country   | Location        | AUC   | Accuracy | F1    | Precision | Recall | N     |
|-----------|-----------------|-------|----------|-------|-----------|--------|-------|
| Palestine | Gaza            | 76.26 | 95.34    | 97.61 | 95.39     | 99.94  | 1783  |
| Ukraine   | Kherson         | 67.61 | 86.53    | 26.8  | 24.53     | 29.55  | 527   |
| Ukraine   | Antonivka       | 70.26 | 90.91    | 38.1  | 66.67     | 26.67  | 143   |
| Ukraine   | Lysychansk      | 75.57 | 75.89    | 85.34 | 76.17     | 97.03  | 419   |
| Ukraine   | Avdiivka        | 63.78 | 79.37    | 88.36 | 79.57     | 99.33  | 189   |
| Ukraine   | Melitopol       | 69.27 | 82.35    | 26.32 | 22.06     | 32.61  | 476   |
| Ukraine   | Sievierodonetsk | 79.41 | 81.42    | 88.4  | 84.83     | 92.27  | 253   |
| Ukraine   | Kramatorsk      | 73.8  | 85.14    | 36.36 | 25.58     | 62.86  | 518   |
| Ukraine   | Mykolaiv        | 71.49 | 79.62    | 28.49 | 20.24     | 48.11  | 1256  |
| Ukraine   | Rubizhne        | 88.4  | 80.65    | 86.88 | 79.89     | 95.21  | 217   |
| Ukraine   | Sumy            | 62.89 | 96.37    | 6.45  | 9.09      | 5.0    | 799   |
| Ukraine   | Shchastia       | 73.86 | 61.76    | 60.61 | 47.62     | 83.33  | 34    |
| Ukraine   | Kremenchuk      | 79.19 | 97.65    | 10.0  | 9.09      | 11.11  | 766   |
| Ukraine   | Kharkiv         | 67.56 | 86.67    | 35.99 | 37.74     | 34.4   | 2588  |
| Ukraine   | Mariupol        | 82.7  | 88.47    | 93.69 | 89.19     | 98.67  | 347   |
| Ukraine   | Borodyanka      | 74.73 | 66.17    | 57.94 | 47.69     | 73.81  | 133   |
| Ukraine   | Volnovakha      | 88.82 | 83.44    | 88.89 | 86.21     | 91.74  | 151   |
| Ukraine   | Irpın           | 86.37 | 79.65    | 87.96 | 79.25     | 98.82  | 226   |
| Ukraine   | Bucha           | 70.92 | 66.18    | 55.13 | 46.24     | 68.25  | 207   |
| Ukraine   | Hostomel        | 74.16 | 71.51    | 82.27 | 70.69     | 98.4   | 186   |
| Ukraine   | Vorzel          | 77.0  | 61.82    | 53.33 | 38.71     | 85.71  | 55    |
| Ukraine   | Moschun         | nan   | 95.0     | 97.44 | 100.0     | 95.0   | 20    |
| Ukraine   | Okhtyrka        | 82.19 | 93.49    | 52.38 | 78.57     | 39.29  | 307   |
| Ukraine   | Trostanets      | 66.33 | 81.22    | 39.29 | 36.67     | 42.31  | 181   |
| Ukraine   | Chernihiv       | 71.1  | 64.78    | 60.15 | 52.67     | 70.11  | 741   |
| Ukraine   | Makariv         | 82.69 | 78.35    | 78.79 | 81.25     | 76.47  | 97    |
| Iraq      | Mosul           | 74.87 | 80.19    | 88.71 | 80.29     | 99.11  | 1141  |
| Sudan     | Yei             | 84.65 | 83.65    | 48.99 | 40.13     | 62.89  | 15345 |
| Syria     | Raqqa           | 92.95 | 92.82    | 96.22 | 93.19     | 99.44  | 195   |
| Syria     | Aleppo          | 71.84 | 77.04    | 86.99 | 77.05     | 99.87  | 993   |
| All       | All             | 84.3  | 83.14    | 65.58 | 71.62     | 60.47  | 30293 |

500m-by-500 m grid-cell level accuracy statistics are generated using UNOSAT damage annotations. For each city, a threshold value of T is selected using a precision–recall curve to generate a binary damage classification. A cell is labeled damaged if it contains one or more damage annotations.

**Table A.10**  
Building-level damage detection accuracy using InSAR coherence change detection.

| Location        | AUC   | F1    | Precision | Recall | N      |
|-----------------|-------|-------|-----------|--------|--------|
| Gaza            | 83.57 | 52.57 | 49.64     | 55.87  | 220148 |
| Lysychansk      | 63.06 | 37.38 | 28.48     | 54.39  | 22280  |
| Sievierodonetsk | 65.64 | 61.86 | 52.92     | 74.45  | 7085   |
| Rubizhne        | 73.56 | 65.28 | 61.24     | 69.88  | 9905   |
| Kharkiv         | 67.16 | 19.02 | 19.51     | 18.55  | 122590 |
| Mariupol        | 72.12 | 72.59 | 59.31     | 93.54  | 20481  |
| Irpın           | 76.53 | 48.95 | 41.0      | 60.72  | 7917   |
| Hostomel        | 84.25 | 73.56 | 71.45     | 75.79  | 4728   |
| Chernihiv       | 74.6  | 38.21 | 33.5      | 44.48  | 33733  |
| Raqqa           | 73.04 | 73.11 | 64.14     | 85.01  | 26728  |
| Mosul           | 68.17 | 35.56 | 33.2      | 38.27  | 151259 |
| Aleppo          | 53.2  | 55.99 | 38.88     | 100.0  | 73634  |
| All             | 62.23 | 37.4  | 41.86     | 33.79  | 700488 |

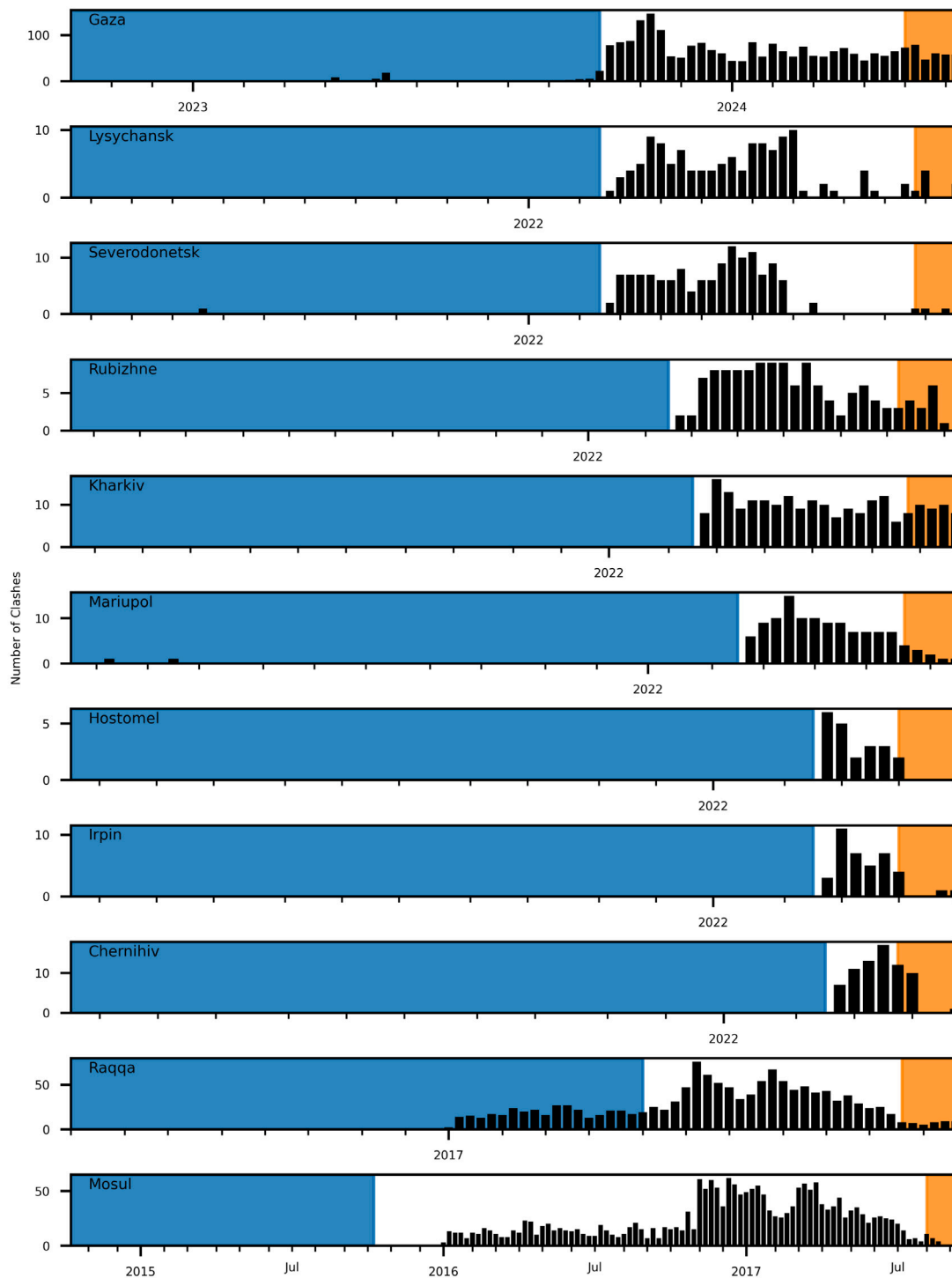
Interferometric Coherence Change Detection was conducted using the last available Sentinel-1 image before the onset of the corresponding conflict (e.g. 2022-02-22 for Ukraine) as a reference image. Coherence change is calculated as the difference between the coherence of the reference image and another pre-war image, with the coherence of the reference image and the first post-war image after the date of the corresponding UNOSAT assessment. For each location, two coherence change detection rasters are averaged together: one from an ascending orbit, and one from a descending orbit, to detect damage that may only be visible from one side. Ukrainian cities with extremely sparse damage (e.g. Sumy, where <0.08% of buildings are damaged) are not included.

Mueller study, inference is conducted at the level of a  $32 \times 32$  meter grid covering each city. The exact AOIs used in their study are not publicly available. Consequently, a hand-drawn AOI matching the yellow bounds shown in Figure 1 of the Mueller study was generated for Aleppo, and for the rest of the cities, the AOIs provided with the UNOSAT damage annotations were employed. Table A.5 compares the number of patches in each city and the proportion thereof that contain at least one destroyed building.

In addition to the cities analyzed in the Mueller study, the PWTT is tested on Damascus and Idlib, neither of which is mentioned in

the previous study. This is an important omission, as Damascus alone contains roughly as many patches as the remaining cities (excluding Aleppo) combined.

The reproduced samples are broadly similar to those used in the Mueller study in terms of the number of patches and the proportion of destruction. Aleppo and Daraa match closely, while the reproduced samples for Hama and Deir-Ez-Zor have a lower proportion of damage, and the reproduced sample for Raqqa had a higher proportion of damage. These discrepancies result from a number of challenges in applying the PWTT to the Syrian context that significantly degrade



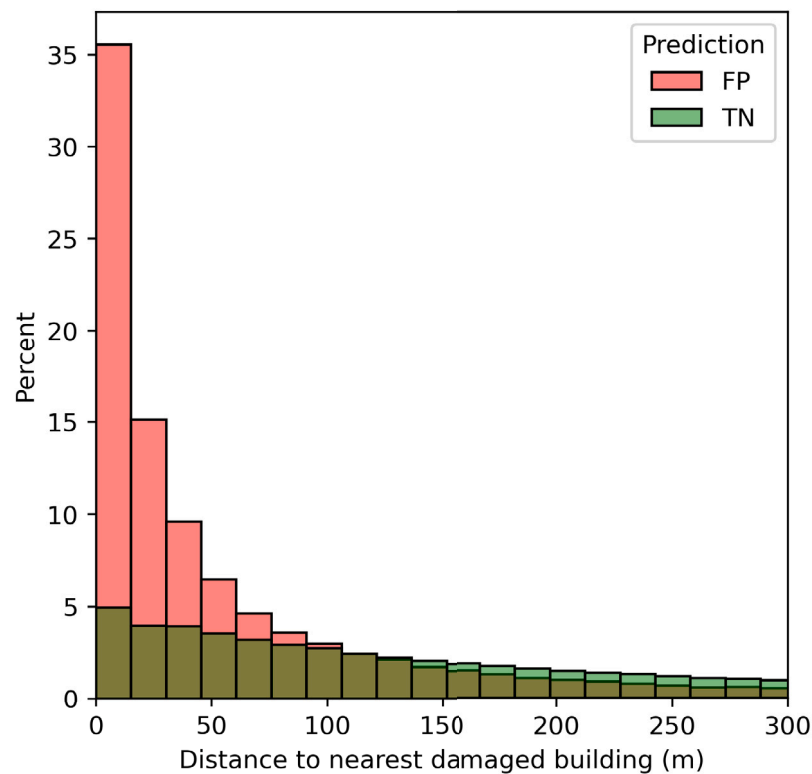
**Fig. A.7.** Reference and inference periods for select cities.

The bars represent the weekly total number of clashes in each city, sourced from the Armed Conflict Location Event (ACLED) dataset. 12 month pre-war reference periods are shaded in blue, spanning one year before the onset of conflict. 1 month post-war inference periods are shaded in orange, beginning with the date of the UNOSAT damage assessment. ACLED data for Syria is only available from 2017 onwards; the subplot for Raqqa is thus missing values during this period, and because the inference period for Aleppo was in 2016, this city was excluded from the plot. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the performance and evaluation thereof, and preclude a one-to-one comparison with the Mueller study.

The reference period for all cities spans from 2014-10-10 to 2015-10-10, reflecting the earliest available Sentinel-1 imagery over the area. As this is nearly 4 years into the war, in all cases significant damage

had already been dealt to each city prior to the availability of Sentinel-1 data, with significant further damage accruing during the reference period. In the case of Homs, this entirely precludes analysis because all recorded damage occurred prior to October 2014. In Hama, the entire district of Masaa Al Arbaeen had been [destroyed by October 2012](#), but



**Fig. A.8.** Distance between false positives and true negatives to the nearest damaged building.

The distance between each False Positive (FP) prediction and the nearest UNOSAT damage annotation is calculated, and the resulting histogram is plotted in red. The same procedure is conducted for True Negative (TN) predictions, and plotted in green. False positives tend to be much closer to damaged buildings than True Negatives, suggesting spillover effects from damage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the date of the damage recorded by UNOSAT for this location is nearly a year later in September 2013.

Where possible, UNOSAT damage annotations preceding the reference period are excluded from the analysis, hence the lower proportion of destroyed patches in Hama and Deir-Ez-Zor compared to the Mueller study. However, many of the damage annotations were first recorded several years after the actual damage, meaning they cannot easily be filtered. For example, Fig. A.6 shows the Jobar neighborhood of Damascus was nearly completely destroyed by August 2014. However, a UNOSAT damage assessment published in 2017 shows 1,226 destroyed buildings in this area, but labels the first recorded observation of damage as June 2016. Naturally, the PWTT fails to identify these as they were damaged before Sentinel-1 imagery was available, and with no way of filtering these observations, they register as false negatives.

Table A.6 below compares the results of the PWTT to the results of the deep learning approach taken by Mueller et al. (2021). AUC and Average Precision calculated on the full unbalanced sample are compared to the results in Table 2, in which training and testing is conducted on the same city, and Table S3 in the appendix, in which a model is trained on Aleppo only. In all cases, the PWTT yields lower AUC and average precision than a deep learning model trained and tested on the same city.

Yet as discussed in Section 1.1, the most realistic real-world scenario is one in which there are no pre-existing damage labels for a given city. Therefore the comparison between the PWTT and the out-of-domain results in the Mueller study is the most relevant. There are four cities for which such a comparison is possible: Raqqa, Daraa, Hama, and Deir-Ez-Zor. Compared to a model trained only in Aleppo, the PWTT outperforms the deep learning approach in Raqqa and Deir-Ez-Zor, but underperforms in Daraa and Hama (see Table A.6).

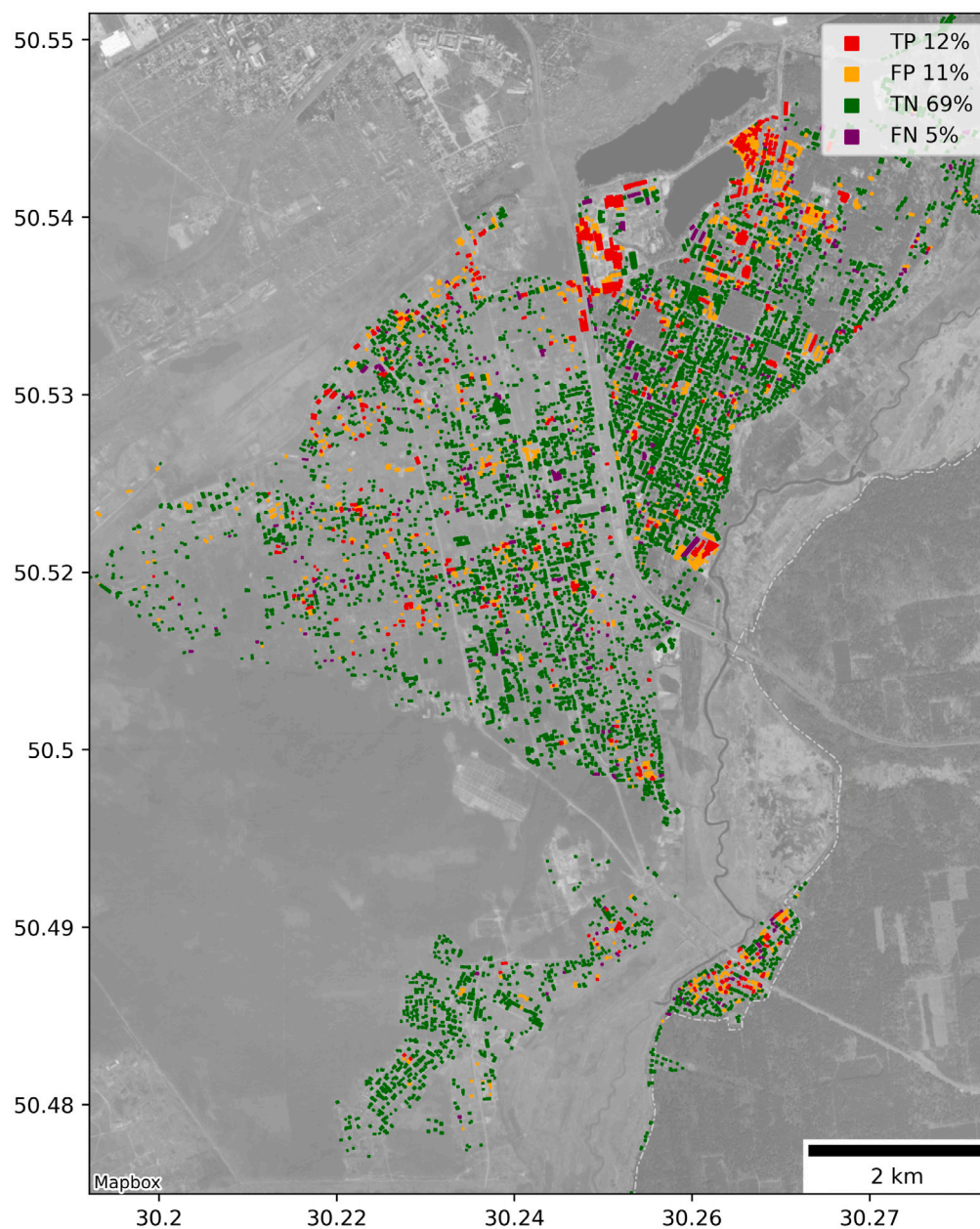
In comparing these results, it is important to consider the differences between the samples reported in Table A.6. However, there is no systematic directional bias in the sample characteristics that aligns with the differences in performance. Compared to the referenced study, the proportion of damaged buildings is identical Daraa, lower in Deir-Ez-Zor and Hama, and higher in Raqqa. The number of patches is nearly the same in Deir-Ez-Zor, lower in Daraa and Hama, and higher in Raqqa. As such, it is not the case that the PWTT only outperforms the deep learning approach in cases where the sample has a higher proportion of damage, or a larger AOI (see Figs. A.9–A.12).

Despite the many challenges of applying the PWTT to the Syrian context – which include only being able to conduct inference nearly 5 years after the start of the war and the inability to exclude damage labels that pre-date the availability of Sentinel-1 imagery – results are competitive with the results reported by Mueller et al. (2021) in out-of-domain scenarios. While the PWTT is less accurate than a deep learning model trained and tested on the same Syrian city, out-of-domain scenarios are an important test of the performance of building damage detection models, as these should not be reliant on manual damage annotations within the same city in order to perform adequately. Thus, the PWTT achieves similar results to one of the leading deep learning approaches in out-of-domain scenarios, using freely available imagery and lightweight statistical change detection.

#### Data availability

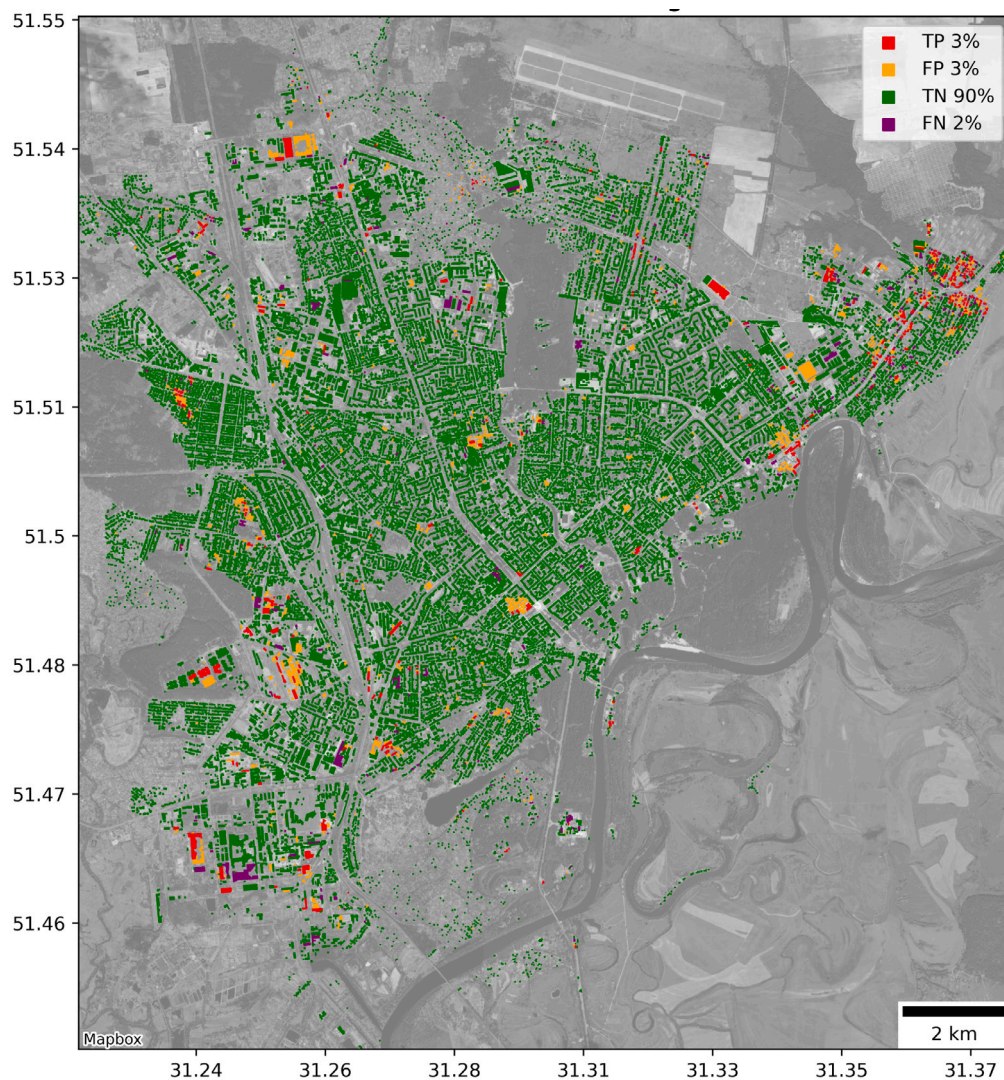
I have linked to the data/code in the Abstract: <https://github.com/oballinger/PWTT>.





**Fig. A.9.** Predicted and observed damage, Irpin.

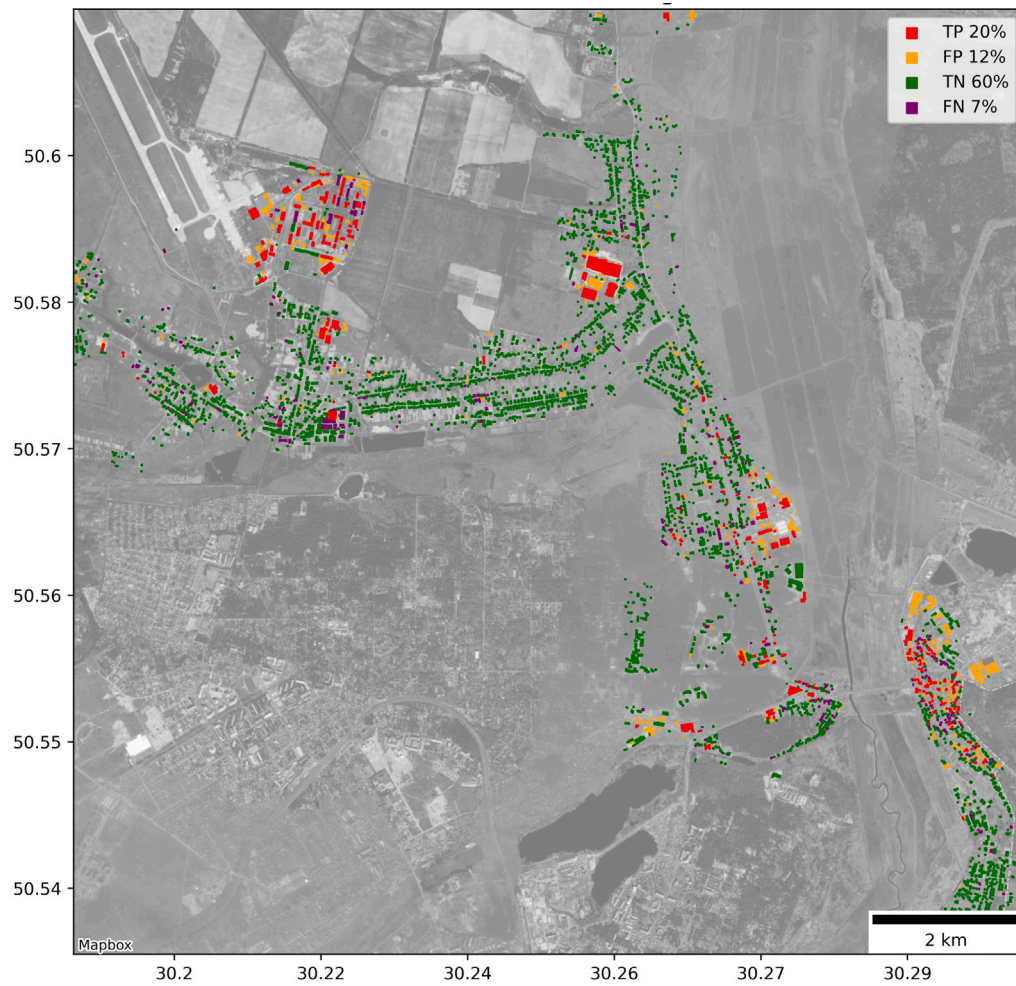
Predicted damage is compared to observed damage in Irpin, Ukraine. True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. A.10.** Predicted and observed damage, Chernihiv.

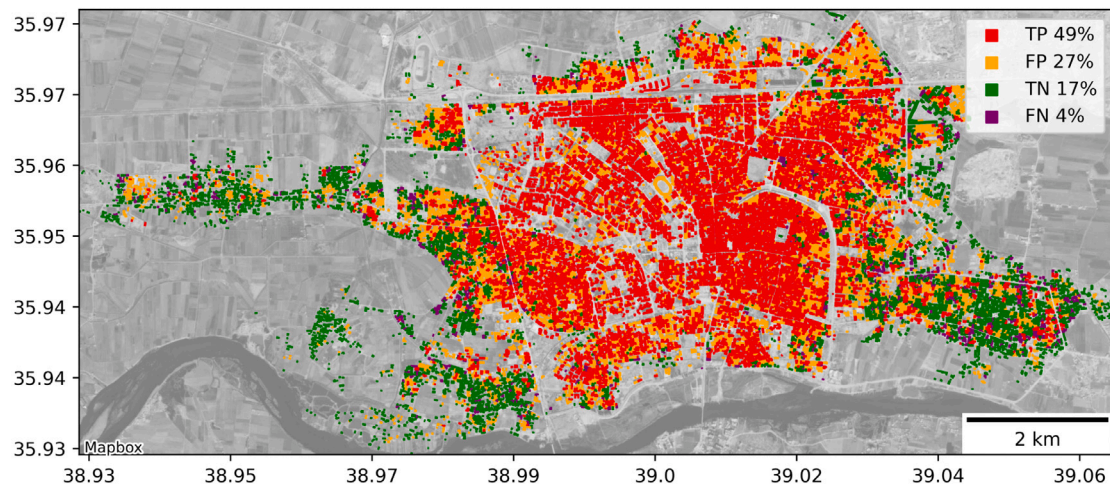
Predicted damage is compared to observed damage in Chernihiv, Ukraine. True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)





**Fig. A.11.** Predicted and observed damage, Hostomel.

Predicted damage is compared to observed damage in Hostomel, Ukraine. True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. A.12.** Predicted and observed damage, Raqqa.

Predicted damage is compared to observed damage in Raqqa, Syria. True Positive and True Negative predictions are shown in red and green, respectively, while False Positives and False Negatives are shown in orange and purple. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



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